

Fundamental Follower Needs

● Protection

Ensure followers remain safe, especially in the face of external threat.

● Affiliation

Ensure followers feel strongly connected and supported.

● Status

Help followers to acquire status and resources.

● Vision

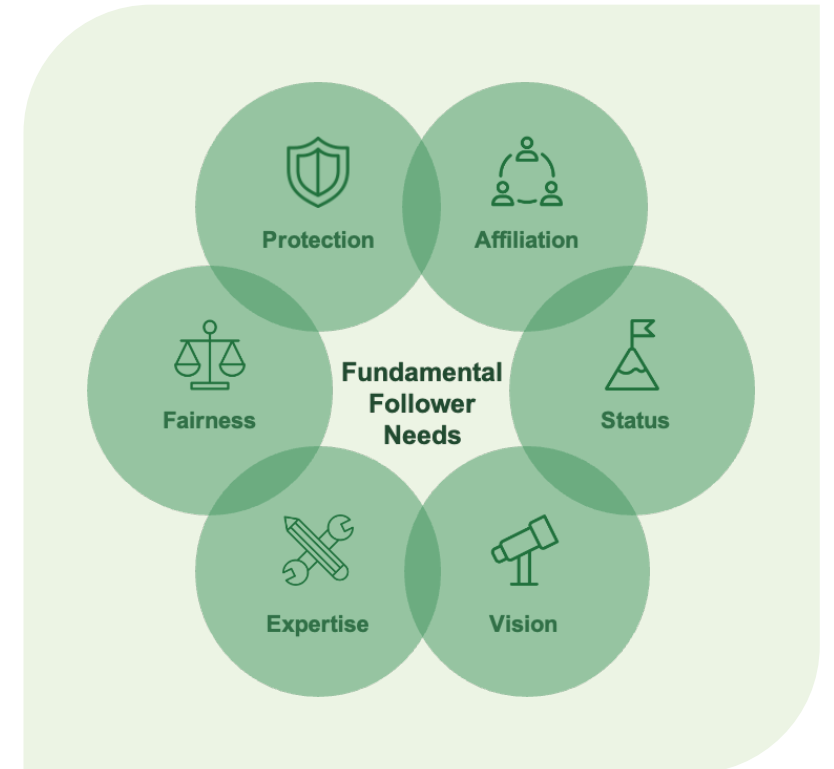
Provide guidance to followers based on their vision.

● Expertise

Provide guidance to followers based on their expertise.

● Fairness

Ensure followers are treated fairly and effectively manage conflicts of interest within the group.



Source:

- Sheng, X., Andrews, W., & van Vugt, M. (2026). The psychology of following: Conceptualizing and validating the Fundamental Follower Needs Inventory. *Journal of Applied Psychology*, 111(6), 768-801. <https://doi.org/10.1037/apl0001347>
- Van Vugt, M., Sheng, X., & Andrews, W. (2026). Are you meeting the needs of the people you lead? *Harvard Business Review*. <https://hbr.org/2026/05/are-you-meeting-the-needs-of-the-people-you-lead>

Fundamental Follower Needs Inventory (FFNI) A 22-item Measurement Tool

Needs	Example Items
Protection	• I wish to have a leader who positions themselves between my group and an outside threat. • I am motivated to follow a leader who stands up for me against outsiders.
Affiliation	• Having a leader who can bring the group together is valuable to me. • It's important for me to have a leader who makes me feel part of the group.
Status	• I need a leader who improves my standing. • I wish to have a leader who helps our group achieve status.
Vision	• I wish to have a leader who sets clear goals. • I am motivated to follow a leader who has a clear vision.
Expertise	• I need a leader who enables me to learn from them. • It's important for me to have a leader who offers advice.
Fairness	• Having a leader who makes fair and balanced decisions is valuable to me. • It's important for me to have a leader who manages within-group conflicts in a just way.

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Note: The full item list and participant instructions are available in the article: Sheng, X., Andrews, W., & van Vugt, M. (2026). The psychology of following: Conceptualizing and validating the Fundamental Follower Needs Inventory. *Journal of Applied Psychology*, 111(6), 768-801. <https://doi.org/10.1037/apl0001347>

The English version of the FFNI is in the main text, and the Chinese version is in the supplementary materials.
Alternatively, you can email Dr. Xiaotian (Shirley) Sheng (shirleyshengxt@gmail.com) to request the full list of items.

**The Psychology of Following: Conceptualizing and Validating the Fundamental Follower
Needs Inventory**

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Abstract

Humans possess an evolved followership psychology that enables them to identify and endorse different types of leaders depending on situational demands. But what fundamental needs guide these follower endorsements? Across a preliminary study and five validation studies ($N = 3,514$), we developed and validated the Fundamental Follower Needs Inventory (FFNI)—a psychometrically robust measure that identifies six core follower needs: *protection*, *affiliation*, *status*, *guidance* (including *vision* and *expertise*), and *fairness*. In Studies 1 and 2, we conducted content validation, tested reliability, and confirmed the factor structure of the FFNI across three domains (general, political, and workplace), three countries (United States, United Kingdom, and China), and multiple time points. Study 3 demonstrated FFNI's convergent and discriminant validity. Studies 4 and 5 explored the nomological network, examining its antecedents, consequences, and both predictive and incremental validity. The FFNI provides a novel tool for researchers to investigate how follower needs vary across contexts and cultures, and how these needs shape leader endorsements and perceptions of leadership effectiveness. Practically, the FFNI offers leaders a framework to better understand and respond to the psychological needs of those they lead.

Keywords: follower needs, motivation, implicit leadership theories, evolutionary psychology, measure development

The Psychology of Following: Conceptualizing and Validating the Fundamental Follower Needs Inventory

Why would a 30-year-old working-class male be more likely to vote for a different political candidate than a 65-year-old middle-class female? Why would employees with distinct personalities—or from different cultural backgrounds—have divergent views about what makes an ideal workplace manager? Although prior research has distinguished different follower types, styles, and strategies (Bastardo & Van Vugt, 2019; Coyle et al., 2023; Goswami et al., 2022; Kellerman, 2007; Kelley, 1988; Van der Velde & Gerpott, 2023), we still know surprisingly little about the deeper psychological forces that lead individuals to identify, support, and follow particular leaders. While it is well established that people endorse different kinds of leaders across different contexts (Aktas et al., 2016; House et al., 2004; Van de Vliert, 2006), we lack a unifying framework to explain why these evaluations exist. Specifically, there is no consensus on the fundamental motives, desires, and needs that guide these endorsements, nor do we possess validated tools to reliably measure them.

The current paper addresses these gaps by taking a functional, evolutionary perspective on leadership and followership (cf. Bastardo & Van Vugt, 2019; Van Vugt & Ronay, 2014) to develop an extended framework of follower needs and use this framework to develop and validate an inventory that can be used to reliably assess fundamental follower needs across different people, contexts, and cultures, as well as explore the downstream consequences for leadership ideals and perceptions of effectiveness. In doing so, we draw on two key foundational models: the Triad Model of Follower Needs (De Waal-Andrews & Van Vugt, 2020), which identifies *protection*, *guidance*, and *fairness* as core follower needs, and the revised hierarchy of needs framework (Kenrick et al., 2010), which highlights additional motives such as *affiliation*

and *status* that leaders can help fulfill through fostering social connection and enabling followers' access to valued resources.

In this paper, we address four interrelated questions. First, why is it important to develop a taxonomy of fundamental follower needs (*Theory*)? Second, what do we regard as a fundamental follower need and which needs can be distinguished from theory (*Conceptualization*)? Third, how can we reliably measure these follower needs (*Assessment*)? Fourth and finally, how do individual (e.g., age and gender) and contextual factors (e.g., national culture) activate these fundamental follower needs differently, and what are the consequences for implicit leadership theories and perceived leadership effectiveness (*Nomological network*)?

We address these questions both theoretically and empirically. We first explain why it is important to develop a taxonomy of fundamental follower needs. Second, we define the construct of a fundamental follower need, and following a systematic process, rooted in evolutionary reasoning, we identify an initial set of needs. Third, in a series of five studies, we create and test an instrument to capture these needs reliably, the Fundamental Follower Needs Inventory (FFNI). Fourth, we test the FFNI across different contexts (work, politics) and cultures (U.S., U.K., China) and provide evidence for its convergent, discriminant, incremental, and predictive validity.

The present research makes theoretical, methodological, and practical contributions to the emerging literature on followership. First, we extend prior theorizing by articulating a taxonomy of fundamental follower needs, rooted in evolutionary perspectives, that links these core needs to the key coordination problems that our ancestors faced, for which leadership and followership emerged as adaptive solutions (e.g., resource acquisition, group defense, collective movement). Second, we create and validate a tool—the Fundamental Follower Needs Inventory (FFNI)—to

assess these fundamental needs, and their unique and joint predictive power in predicting implicit leadership theories and leader effectiveness across individuals, contexts, and cultures. Third, the FFNI can be potentially used for a range of practical applications, from the contents of leadership development programs to the predictions of political elections.

Why We Need a Taxonomy of Follower Needs?

Despite a surge of interest in followership, most existing models still lack a unifying framework that classifies what followers actually need from leaders (Cornelis et al., 2013; Pierro et al., 2014; Tepper et al., 2018; Yun et al., 2006). We propose that an evolutionary perspective supplies that missing foundation and yields at least four distinct advantages. First, evolution offers a functional rationale for identifying and classifying fundamental follower needs. Leadership and followership co-evolved as adaptive solutions to recurring coordination problems in ancestral groups, such as defending territory, acquiring resources, and making collective decisions about group movement and sharing resources (Van Vugt et al., 2008; Von Rueden & Van Vugt, 2015). From this vantage point, core follower needs for such services as protection, expertise and guidance by leaders can be viewed as domain-specific adaptations that historically enhanced individual and group survival. An evolutionary taxonomy therefore separates ultimate (functional) explanations—this leader can protect us—from proximate (psychological) mechanisms—this leader is charismatic—providing a principled way to decide which needs belong in the “fundamental” set and why those needs should generalize across cultures and contexts.

Second, it moves the field beyond descriptive typologies to motivational drivers. In Kelley’s (1988) and Kellerman’s (2007, 2008) seminal theoretical contributions on followership, they proposed different typologies of followers. Kelley (1988) distinguished between more active and

passive followers, as well as more and less critical followers. Kellerman (2007, 2008) categorized followers as isolates, bystanders, participants, activists, and diehards, with their level of engagement with the leader being the defining criterion. Both typologies adopt a role-based approach, whereby followers take on a subordinate role in their relationship with the more powerful leader. However, there is growing recognition that followers actively shape both the type of leadership that arises and the behaviors leaders display (Bastardo & Adriaensen, 2023; Uhl-Bien et al., 2014). While existing typologies and styles (Bastardo & Van Vugt, 2019; Coyle et al., 2023; Goswami et al., 2022; Kellerman, 2007; Kelley, 1988; Van der Velde & Gerpott, 2023) describe how followers act, they rarely account for why certain leader traits and behaviors become appealing under particular circumstances. By grounding follower needs in evolved motivational systems, our framework uncovers the psychological engines that make leadership possible, thereby clarifying the fundamental interdependence of leadership and followership.

Third, it escapes the “do-good/don’t-do-bad” trap in leadership research. Much of the literature assumes that “positive” leadership styles are universally effective and “negative” styles are universally harmful (Fischer & Sitkin, 2023). An evolutionary analysis shows that followers sometimes prefer dominant leaders when threats loom.¹ For example, authoritarian leadership can evoke favorable reactions when followers’ need for protection is salient—whether in certain cultural contexts (e.g., many East-Asian societies; Wang et al., 2023), in harsh ecologies (Laustsen et al., 2025; Lonati & Van Vugt, 2024), or during acute disruptions (Zhu et al., 2024). Recognizing the situational activation of different needs yields more nuanced predictions than a simple good-versus-bad dichotomy.

¹ While such preferences can be understandable from a functional and evolutionary perspective, it is important to note that the enactment of “negative” leadership still often can come at considerable costs to followers, organizations, and even the leaders themselves.

Fourth, it integrates a fragmented literature into a coherent model. Isolated studies have examined single follower needs (Pierro et al., 2014; Tepper et al., 2018; Yun et al., 2006), such as the need for belonging (Cornelis et al., 2013). Yet without an overarching taxonomy, these findings risk conflating related constructs (e.g., belongingness vs. affiliation vs. fairness). A synthetic model clarifies each need's unique incremental effect, helping researchers avoid redundancy and uncover boundary conditions.

An evolutionary taxonomy of fundamental follower needs provides the theoretical glue that followership research currently lacks. It explains why certain leader behaviors resonate, predicts when ostensibly negative styles may be functional, distinguishes leader-specific needs from broader work needs, and unifies scattered findings under a single framework. In doing so, it shifts the analytic lens from leader-centric to follower-driven dynamics—a critical step for both theory and practice in modern organizations.

What Are Fundamental Follower Needs?

So how do we define, identify, and operationalize a fundamental follower need? We followed the steps recommended by Podsakoff et al. (2016). We first consulted dictionary definitions (Cambridge University Press, n.d.; Merriam-Webster, n.d.; Oxford University Press, n.d.) that describe “fundamental” as “central”, “generative”, and “innate”. Second, we surveyed the literature to identify a set of related constructs, their key attributes, and how these key attributes map onto our definition of fundamental follower needs (see Table S1 in the Supplemental Materials and the subsequent Differentiating FFNs From Related Constructs section). As for theoretical grounding, inspired by evolutionary psychology, we define a fundamental follower need (FFN) as *an evolutionarily shaped motivation that drives individuals to identify, choose, and support particular leaders in order to solve recurrent coordination*

problems that affect their own and their group's survival and well-being.

Four elements are integral to this definition. First, each FFN has an *evolutionary foundation*. Each FFN is rooted in selection pressures encountered by our ancestors, forming part of an adaptive followership psychology. Second, each FFN has a *functional purpose*. An FFN helps followers detect suitable leaders, choose optimally, and coordinate effectively. The function always maps onto a specific ancestral challenge (e.g., threat defence, resource allocation). Third, each FFN can have *dual-level relevance*. The same need may be triggered by problems that endanger the individual, the group, or, usually, both. A fourth element of an FFN is that it is *follower-centric* in focus. FFNs shape followers' cognitions, perceptions, and behaviours, positioning followership as an active rather than passive process.

Finally, a motive qualifies as an FFN only if it meets the following three criteria. First, the need is reliably aroused by modern contexts that mimic ancestral coordination challenges (*evolutionary foundation*; Bastardo & Van Vugt, 2019; De Waal-Andrews & Van Vugt, 2020; Kenrick et al., 2010). Second, each FFN has predictable outputs. Activation elicits a consistent pattern of outcomes (*functional purpose*; Anderson et al., 2015; Kenrick et al., 2010), notably specific leader ideals and evaluations of leader effectiveness. Third, the FFNs should be human universals, found across different contexts and cultures, but perhaps with variable strength. The need appears across domains (work, politics), cultures (Western and non-Western), and historical periods (Anderson et al., 2015), although its intensity may differ by age, gender, culture, or situational cues (Lonati & Van Vugt, 2024).

Searching for Fundamental Follower Needs

Building on our conceptual definition, we identified FFNs by mapping them onto the recurrent coordination problems human groups faced throughout evolutionary history, such as

defending against threats, acquiring resources, maintaining cohesion, and resolving internal conflict (Bastardo & Van Vugt, 2019; Del Giudice, 2023; Kenrick et al., 2010; Neel et al., 2016; Sheng et al., 2024). To construct a comprehensive and evolutionarily grounded taxonomy of FFNs, we drew on two key theoretical frameworks.

The first is the Triad Model of Follower Needs (De Waal-Andrews & Van Vugt, 2020), which draws on evolutionary leadership theory, positing that humans possess an adaptive followership psychology shaped by selection pressures in ancestral environments (Bastardo & Van Vugt, 2019; Laustsen, 2021; Van Vugt & Ronay, 2014). This model identifies three core FFNs: *protection* against threat, *guidance* toward shared goals, and *fair* dispute resolution, each reflecting a fundamental challenge that leaders historically help address. For example, an external threat, such as conflict with a rival group, activates the need for protection, which in turn increases the support for dominant and (outwardly) aggressive leaders capable of addressing the danger. Empirical studies, including cross-cultural research, have provided robust evidence for such mechanisms (Laustsen & Petersen, 2017; Laustsen et al., 2025; Sheng et al., 2024; Spisak et al., 2012).

Second, to complement this initial set, we also turned to the revised hierarchy of needs developed by Kenrick et al. (2010), which integrates evolutionary insights into Maslow's classic hierarchy-of-needs pyramid (Bridgman et al., 2019). Their renovated pyramid identifies seven fundamental social motives reflecting the adaptive problems our ancestors faced. Among these, three align closely with our definition of FFNs: *self-protection*, *affiliation*, and *status*. These needs are particularly relevant in leader-follower relations, as they correspond to followers' desire for safety, social connection, and access to valued resources.

In defining the boundaries of the FFN construct, we also considered several well-known

needs that, despite their theoretical appeal, do not qualify as fundamental follower needs. For instance, autonomy, a basic psychological need in self-determination theory (Deci & Ryan, 2000; Ryan & Deci, 2000), is conceptually opposed to the need for leadership and thus falls outside the FFN domain. Similarly, self-actualization—despite its iconic role in Maslow’s original pyramid (Peterson & Park, 2010)—reflects self-gratifying goals rather than adaptive functions related to leadership or coordination. Although the need for structure (Neuberg & Newsom, 1993) might seem relevant, it lacks direct ties to survival or reproductive fitness. However, aspects of it may be subsumed under a broader need for guidance, which can encompass more immediate cognitive needs such as the need for closure, structure, and growth.

After synthesizing the insights from both frameworks, we initially constructed a set of five FFNs: *protection*, *affiliation*, *status*, *guidance*, and *fairness*. These needs were chosen for their relevance to leader–follower dynamics across domains (e.g., business, politics) and levels of analysis (individual and group), and we carefully crafted short, generalizable labels for each need. Each FFN in our framework addresses a distinct recurrent adaptive problem in human evolutionary history: *protection* maps onto threats to physical or social safety; *affiliation* supports group cohesion; *status* relates to access to valued resources; *guidance* addresses direction and coordination; and *fairness* facilitates conflict resolution (see Table 1 for an overview). These needs are defined and elaborated in detail in the section below (definitions are also summarized in Table 3).

Identifying the Fundamental Follower Needs

The FFN for *protection* is defined as the need for leaders who ensure (groups of) followers remain safe, especially in the face of external threat. The protection need becomes salient in the face of danger, like wars, pandemics, or other types of immediate threats, leading followers to

support leaders perceived as strong, dominant, and assertive. Empirical studies have shown that during crises, such as intergroup conflict, support increases for leaders who embody strength and masculinity (Laustsen & Petersen, 2017, 2020; Laustsen et al., 2025; McCann, 1997; Murray et al., 2013; Spisak et al., 2012). Winston Churchill is a classic leadership example, rising to prominence during WWII as a symbol of national resilience and wartime decisiveness. This follower need is often triggered by emotional responses such as fear and anger (Ambroziak et al., 2022; Sheng et al., 2024), which signal (perceived) threats to safety.

The FFN for *affiliation* refers to the need for leaders who ensure (groups of) followers feel strongly connected and supported. This need reflects a basic human motivation for belongingness (Baumeister & Leary, 1995). As a social species, humans evolved to depend on group affiliation for survival. Leaders who create a sense of shared identity, community, and empathy are more likely to attract and retain followers. In the business world, Microsoft CEO Satya Nadella has often been recognized for fostering such an affiliative culture, emphasizing empathy and collaboration as core values; he said, “Empathy is not a soft skill. In fact, it’s the hardest skill we learn—to relate to the world, to relate to people that matter the most to us”. Similarly, Queen Elizabeth, in her first Christmas Broadcast in 1952, exemplified this by emphasizing a collective identity— “we belong, all of us, to a far larger family.” Empirical research supports this. Employees high in affiliation needs tend to prefer relational, socially oriented managers (Boatwright & Forrest, 2000), and remote workers increasingly value leaders who cultivate group identity (Shi et al., 2024).

The FFN for *status* is defined as the need for leaders who help (groups of) followers to acquire status and resources. When followers feel their individual or group status is under threat, they often gravitate toward dominant, status-restoring leaders. For instance, Donald Trump’s

“Make America Great Again” campaign resonated with individuals who perceived America’s global standing—and their own social status—as eroding. Research suggests that Trump’s appeal was especially strong among traditionally high-status groups (e.g., white, male, Christian Americans), driven by perceived status loss (Kakkar & Sivanathan, 2017; Mutz, 2018). Some followers may even support aggressive leadership styles as a way to reassert dominance over perceived out-groups (Womick et al., 2019).

The FFN for *guidance* is defined as the need for leaders who provide guidance to (groups of) followers based on their vision and expertise.² This need becomes salient when resources are scarce or available unpredictably (Sng et al., 2018). Followers tend to value leaders with a clear strategic vision (De Waal-Andrews & Van Vugt, 2020) and demonstrated competence (Kenrick et al., 2010). Vision refers to goal-oriented direction, while expertise refers to specialized knowledge and skills. Deng Xiaoping’s leadership during China’s economic transformation and Steve Jobs’s influence at Apple both reflect this duality. Deng mobilized national reform through a unifying vision, while Jobs combined visionary thinking with technical expertise in design and marketing. This FFN might be closely linked to endorsements for charismatic or transformational leaders (Podsakoff et al., 1990; Shamir & Howell, 1999) and for leaders seen as prestigious and highly competent (Cheng et al., 2013; Van Vugt & Smith, 2019).

Finally, the FFN for *fairness* refers to the need for leaders who ensure (groups of) followers are treated fairly and who effectively manage conflicts of interest within the group. Followers want leaders who act with integrity, impartiality, and a commitment to justice (De Waal-Andrews & Van Vugt, 2020). Nelson Mandela stands as an exemplar; his leadership in post-

² During the validation process (see Study 2), we further differentiated the broader dimension of *guidance* into two factors: *vision*-based guidance and *expertise*-based guidance. Specifically, the FFN for *vision* refers to the need for leaders who provide guidance to (groups of) followers based on their vision, whereas the FFN for *expertise* refers to the need for leaders who provide guidance to (groups of) followers based on their expertise.

apartheid South Africa was defined by non-racial democracy and a strong moral compass. This need underpins support for ethical leadership (Brown et al., 2005), but interestingly, it may also motivate support for dominant leaders when fairness is enforced through discipline and punishment (Chen et al., 2021; De Waal-Andrews & Van Vugt, 2020). In contexts where group cooperation is fragile, strong leadership may be valued for its ability to deter freeloading and enforce norms.

Together, these five FFNs—*protection*, *affiliation*, *status*, *guidance*, and *fairness*—offer a functional and evolutionarily grounded framework for understanding how different follower needs shape leadership evaluations across diverse contexts. In Study 2, we will examine this multidimensional structure of FFNI to demonstrate its psychometric properties across domains, countries, and times.

Differentiating FFNs From Related Constructs

To support the convergent and discriminant validation of our framework and inventory (see also Study 3), we should differentiate FFNs from several related constructs. As elaborated below, we anticipate that FFNs will exhibit meaningful correlations with the need for leadership, fundamental social motives, and basic personal values, while remaining empirically distinct from them. By contrast, we do not expect FFNs to correlate with basic psychological needs satisfaction and frustration.

Need for Leadership

Need for leadership refers to an individual's general desire for leaders to facilitate goal attainment in the workplace (De Vries et al., 2002). While related, this construct differs from FFNs in two key ways. First, it is unidimensional, reflecting an overall desire for leadership, whereas FFNs are multidimensional, capturing distinct types of needs (e.g., for *protection*,

status, fairness). Second, the need for leadership measure is domain-specific—focused on workplace leaders only—whereas FFNs apply across multiple domains, including politics, family, and organizational life. Despite these differences, both constructs reflect expectations from leaders, and we therefore anticipate positive correlations between all FFNs and the general need for leadership.

Fundamental Social Motives

Fundamental social motives are evolutionarily shaped motivational systems designed to address recurring social challenges (Neel et al., 2016). These include self-protection, disease avoidance, affiliation, status, mate seeking, mate retention, and kin care. Like FFNs, they reflect adaptations, but they are broader: leadership plays a limited role in fulfilling motives like mate retention or kin care. Moreover, FFNs include unique elements, such as the needs for *guidance* and *fairness*, which are absent from the social motive framework. We expect FFNs for *protection, affiliation, and status* to be positively related to the corresponding fundamental social motives (e.g., self-protection/disease avoidance, affiliation, and status), while others may not align directly.

Basic Personal Values

Basic personal values refer to trans-situational goals that guide attitudes and behaviors across life contexts (Schwartz, 1992; Schwartz et al., 2012). Although FFNs and personal values both involve motivational content, they differ in focus. Values reflect beliefs about desirable life goals (e.g., security, benevolence, achievement), while FFNs reflect specific expectations about desirable leader behaviors. Despite this distinction, we anticipate meaningful associations: the FFN for *protection* may relate to the value of security, *affiliation* to benevolence, *status* to power/achievement, *guidance* to self-direction, and *fairness* to universalism.

Basic Psychological Needs Satisfaction and Frustration

Self-determination theory (SDT) identifies three basic psychological needs—autonomy, competence, and relatedness—that, when satisfied, enhance well-being (Deci & Ryan, 2000; Ryan & Deci, 2000; Van den Broeck et al., 2016). Although both SDT and FFNs focus on individual needs, they differ in emphasis. SDT measures experienced satisfaction or frustration, whereas FFNs assess desired qualities or expectations from leaders. For example, someone may desire affiliation from a leader (FFN) without feeling particularly satisfied or frustrated in their broader relatedness at work (SDT). As such, needs and experiences are not always aligned, and we do not expect strong or specific relationships between FFNs and need satisfaction/frustration.

What Are the Antecedents and Outcomes of Fundamental Follower Needs?

Having defined the core set of fundamental follower needs (FFNs), we now turn to their antecedents and outcomes. This section outlines the theoretical nomological network used in Studies 4 and 5 to validate the construct, focusing on where (e.g., domain, country), who (e.g., age, gender), and how (e.g., cognition and perception) these needs operate. We conceptualize FFNs as possessing both trait-like stability and state-like flexibility. On the one hand, FFNs should show stability across different leadership domains (e.g., work and politics), countries (e.g., Western and non-Western societies), and individual traits such as age and gender. On the other hand, they are sensitive to environmental cues—such as wars, recessions, or technological disruptions—that can temporarily heighten a specific follower need. Figure 1 presents our full theoretical and empirical nomological network.

Antecedents

In this study, we investigate how fundamental follower needs differ across leadership domains (work vs. politics), countries (the United States, the United Kingdom, and China), and

demographic groups defined by age and gender. These factors correspond to two central dimensions of our theoretical framework: the “*where*” and the “*who*”. Specifically, the leadership domain and country serve as contextual antecedents (“*where*”), while age and gender represent individual antecedents (“*who*”). We include leadership domain because most prior research on the antecedents of leader preferences has concentrated on national or political leadership (e.g., the psychological needs driving support for dominant political figures), whereas followers in workplace contexts may exhibit distinct needs that vary with their physical and social proximity to leaders, as well as the frequency of their interactions (see leader distance theory; Antonakis & Atwater, 2002). Cross-national comparisons are included because countries differ in their behavioral ecological conditions, reflecting historical and ecological challenges that shape enduring variations in follower motivations. Finally, age and gender are examined as key individual differences, given that life history theory identifies them as fundamental determinants of social motives across the human lifespan (Neel et al., 2016).

Although prior theorizing and scattered empirical evidence hint that these antecedents may influence FFNs, the current body of knowledge remains too limited to justify firm directional hypotheses for each need. Therefore, we broadly anticipate that *FFNs will differ across leadership domains, countries, age, and gender*, but we treat the exact pattern of these differences as an open empirical question to be addressed (Study 4). At the same time, where existing theory and research provide stronger guidance, the following sections outline more specific expectations and propose illustrative patterns of variation that merit empirical testing.

First, follower needs are shaped by the domain in which leadership occurs. *Protection* needs, for example, are likely to be more pronounced in political settings that involve threats to national security than in workplace contexts, where danger cues tend to be more subdued. Both

contexts may activate *protection* needs, but political environments are expected to do so more intensely.

The broader ecological context of a country also plays a role. From a behavioral ecological perspective (Sng et al., 2018), local environmental conditions influence which follower needs are most salient. In high-density societies such as China, where resource competition is fierce, *status* becomes a particularly important need, functioning as a marker of access to resources and opportunities for advancement. Consequently, cross-national differences in the relative prominence of specific FFNs are likely to emerge.

Age, as conceptualized by life history theory (Del Giudice et al., 2016; Kaplan & Gangestad, 2015), is another key factor shaping FFNs. Younger individuals, still navigating status hierarchies and establishing their social positions, are likely to place greater emphasis on *status*-related needs. In contrast, older adults—drawing on accumulated experience and social capital—may be less dependent on leaders for *guidance*. Such age-related patterns reflect broader shifts in fundamental social motives across the lifespan (Neel et al., 2016).

Gender differences in FFNs are also theoretically expected. Because men tend to possess greater physical formidability (Neel et al., 2016), they may have less need for leadership as a source of *protection*. Women, by contrast, may show heightened sensitivity to social and physical threats, increasing the salience of *protection* needs. Moreover, sexual selection theory (Trivers, 1972) suggests that men's stronger intrasexual competition makes *status* a particularly central need, given its role in securing mates and social influence. Thus, while both sexes face similar adaptive challenges, evolutionary pressures have shaped systematically different constellations of follower needs.

Outcomes

In the present research, we examine two central outcomes of FFNs: implicit leadership theories (ILT) as *cognitive* outcomes and perceived leadership effectiveness as *perceptual* outcomes. Regarding ILTs, *we expect that distinct FFNs will systematically align with different leadership prototypes, though the precise pattern of these associations remains to be determined empirically*. In contrast, for perceived leadership effectiveness, we specifically hypothesize that *each FFN will be positively related to the perceived effectiveness of leaders who satisfy that particular need* (see Study 5 and Table 14 for detailed hypotheses).

ILTs are followers' mental schemas about what makes an ideal leader (Lord et al., 2020; Offermann & Coats, 2018). Prior research shows ILTs vary by domain (Lord et al., 1984; Metz, 2021), culture (House et al., 2004), and demographics (Epitropaki & Martin, 2004; Mirowska et al., 2022). We propose that FFNs explain these differences by shaping which leader traits followers view as ideal. For example, individuals high in *status* needs may see dominance as prototypical, while those high in *affiliation* needs may favor warmth and sensitivity.

Similarly, FFNs influence perceived leadership effectiveness—followers' subjective judgments of a leader's performance. Research shows ILTs shape these evaluations (Bass & Avolio, 1989; Lord et al., 2020), and we argue that FFNs operate in the same way: followers are more likely to judge a leader as effective when their style aligns with their salient needs. Though this relationship has not been directly tested, related findings suggest that congruence between follower needs and leadership style enhances outcomes like commitment and performance (De Vries et al., 2002).

To examine these associations, we focus on three theoretically relevant leadership frameworks. Transformational leadership is the dominant paradigm in leadership research (Podsakoff et al., 1990; Wang & Howell, 2010), paternalistic leadership is especially salient in

East Asian cultures (Cheng et al., 2014; Cheng et al., 2004), and the status attainment framework offers an evolutionarily grounded approach aligned with FFNs (Bai et al., 2020). Table 14 presents our pre-registered hypotheses outlining how specific FFNs predict perceived effectiveness within each framework.

How to Measure Fundamental Follower Needs?

Overview of the Present Research

Table 2 shows the overview of study phases and samples. We first use our taxonomy of fundamental follower needs to develop a preliminary item pool and refine it to the 22-item Fundamental Follower Needs Inventory (FFNI; Preliminary study). We test if these items accurately reflect the different dimensions of the fundamental follower needs, thereby providing evidence for content validity (Study 1). We then analyze five independent samples to evaluate psychometric properties of the FFNI across different leadership domains (generic, work, and politics), countries (U.S., U.K., and China), and times (two waves; Study 2). Next, we examine the associations of fundamental follower needs with related and unrelated constructs, thus confirming both convergent and discriminant validity (Study 3). Within its nomological network, we also explore fundamental follower needs' antecedents (e.g., domains, countries, age, and gender; Study 4) and outcomes (e.g., implicit leadership theories and perceived leadership effectiveness; Study 5).

Transparency and Openness

We adhered to the *Journal of Applied Psychology* methodological checklist. This study, “The Follower Needs Scale Development and Validation,” received approval from the Scientific and Ethical Review Board (VCWE) of the Faculty of Behavior & Movement Sciences, Vrije University Amsterdam (No. VCWE-2022-166). The sources of all the measures used in our

studies are included in the Appendix. All data, analysis code, and survey materials are available upon request from the author team. Data were analyzed using Jamovi, Version 2.3 (The jamovi project, 2022), which is built on top of the R statistical language, Version 4.1 (R Core Team, 2021). The Chinese sample study's (Study 5; Sample E) design, hypotheses, and analysis plan were preregistered; see <https://aspredicted.org/vt3x-j9vb.pdf>. Other studies were not preregistered because the establishment of factor structure is partially emergent.

Preliminary Study: Item Generation, Selection, and Refinement (Version 0.1 to 1.0)

The main goals of this preliminary study were first to generate the initial item pool, constitute Version 0.1 of the FFNI (FFNI v0.1), and then to conceptually and empirically select and refine the items of FFNI v0.1 to get the final items of FFNI v1.0.

First, we drafted 12 items per need to operationalize the fundamental follower needs. These 60 items constituted the initial item pool FFNI v0.1. We considered three aspects when generating the items: appropriate number, structure, and content (Clark & Watson, 1995). For the number of items, we aimed to ideally retain six items per need or at least three items per need, based on the review of organizational psychology scales (Colquitt et al., 2019, p. 1257; Number of items: $M = 4.79$, $SD = 1.67$, $n = 112$ scales). Thus, we wrote 12 items for each need to ensure adequate items, anticipating that about a quarter to half would survive the iterative validation process (see the systematic review results from Morgado et al., 2017, p. 9, 12). For the structure of items, we drafted six stems to give variety to the items: "I need a leader who [...]," "I wish to have a leader who [...]," "I am motivated to follow a leader who [...]," "Having a leader who [...] is valuable to me," "It's important for me to have a leader who [...]," and "Having a leader who [...] is unimportant to me (reverse-worded)".³ For the content of items, we adopted a

³ The reverse-worded items were dropped based on experts' feedback and poor content validity in our preliminary study. As a reference, only five scales (4%) contained reverse-worded items across the 112 new scales published in

deductive approach to generate items covering the definition of each fundamental follower need. For instance, we drafted items for *protection* considering both individual (e.g., “I am motivated to follow a leader who stands up for me against outsiders”) and collective aspects (e.g., “I wish to have a leader who positions themselves between my group and an outside threat”). Also, we generated items for *guidance* considering both abstract vision (e.g., “Having a leader who moves the group towards a shared direction is valuable to me”) and concrete expertise aspects (e.g., “I need a leader who enables me to learn from them”).

Second, to select and refine the items of FFNI v0.1 to get the final items of FFNI v1.0, we followed the recommended steps for construct development and validation (Lambert & Newman, 2023). In a nutshell, we used eight iterative data collections (total $N = 952$) to (1) seek feedback and suggestions from experts, (2) test the content validity, (3) test the measurement model with confirmatory and exploratory factor analyses (CFA/EFA), (4) re-test the content validity, and (5) re-test the measurement model with CFA. For detailed information on each data collection (e.g., purposes, participants, procedure, and results), see S2 in the Supplemental Materials (SM; Tables S2.1-S2.10). Finally, we achieved the final 22-item FFNI v1.0: four items for *protection*, *affiliation*, *status*, and *fairness*, and six items for *guidance*.

Study 1: Content Validation

The purpose of Study 1 was to confirm the content validity of the 22-item FFNI based on two core features: definitional correspondence and definitional distinctiveness (Colquitt et al., 2019). Definitional correspondence refers to the degree to which the items correspond to the construct’s definition. Definitional distinctiveness refers to the degree to which the items

the *Journal of Applied Psychology*, *Academy of Management Journal*, *Personnel Psychology*, and *Organizational Behavior and Human Decision Processes* from 2010 to 2016. In addition, the scale’s content validity tends to be worse if using reversed items (Colquitt et al., 2019, p. 1257). Based on the above, we determined not to use reverse-worded items.

correspond stronger to the focal construct's definition than to the orbiting constructs' definitions. For instance, the content of a *protection* item (e.g., *I am motivated to follow a leader who stands up for me against outsiders*) should sufficiently correspond to the definition of the follower's need for protection (i.e., definitional correspondence) and correspond much weaker to the definitions of the follower needs for *affiliation*, *status*, *guidance*, and *fairness* (i.e., definitional distinctiveness). We generally expected that the 22-item FFNI would establish both strong correspondence and distinctiveness.

Participants and Procedure

We recruited 210 U.S. adults through *Prolific* (Douglas et al., 2023; Palan & Schitter, 2018) based on five prescreen requirements: 18 years or older, United States residence and nationality, English as the first language, did not participate in our preliminary study, and needed to have earned at least a 99% approval in the system. We paid the participants £1 (around \$1.21) for their participation in this 10-minute survey. To ensure data quality, participants were excluded from the analysis if they failed to pass both the careless respondent questions, in which they were instructed to choose a particular rating (Huang et al., 2012; Meade & Craig, 2012). The final sample of 205 participants consisted of 51.2% females and ranged in age from 19 to 78 years old ($M = 40.61$, $SD = 14.27$).

We utilized Hinkin and Tracey (1999)'s classic rating approach to establish content validity, following Colquitt et al. (2019)'s detailed and replicable instructions. We randomly assigned participants into two groups ($N_{\text{Group1}} = 103$, $N_{\text{Group2}} = 102$). After obtaining informed consent from the participants, we first presented the definition of one of the fundamental follower needs at the top of the page as the focal construct (e.g., the definition of the follower need for *protection*). Then, we listed items from the focal constructs and two different orbiting constructs within each

group (e.g., in addition to listing items from *protection* in both groups, we listed items from *affiliation*, and *status* in Group 1, whereas we listed items from *guidance*, and *fairness* in Group 2; see Table 3). This design enabled each need to be the focal construct, and the other four to act as the orbiting constructs.

Participants' task was to assess the degree to which each item listed on that page matched the concept definition provided, using a 7-point scale (1 = *Item does an extremely bad job of measuring the bolded concept above*, 7 = *Item does an extremely good job of measuring the bolded concept above*). Before the formal task, participants had to pass a practice trial to help them understand and become familiar with the task. In the formal task, each participant rated 66 items in total, separately on five pages (corresponding to the definitions of the fundamental follower needs), and each page contained 12 to 14 items (for detailed patterns, see Table 3).

Results

Table 3 presents the content validation statistics for each of the fundamental follower needs. The 22 items obtained an average definitional correspondence rating of 6.24 on the focal constructs, whereas they only had an average definitional correspondence rating of 3.43 on the orbiting constructs. Based on these informative statistics, we evaluated the content validity by calculating two indices. First, to evaluate the definitional correspondence, we generated the index of *Hinkin Tracy Correspondence* (*htc*)—dividing 6.24 by 7 (seven scale anchors), which equaled .89. Second, to evaluate the definitional distinctiveness, we generated the index of *Hinkin Tracy Distinctiveness* (*htd*)—dividing (6.24 minus 3.43) by 6 (seven scale anchors minus one), which equaled .47. These two content validation indexes confirmed that the FFNI items were equipped with both strong correspondence and distinctiveness, according to the evaluation criteria recommended by Colquitt et al. (2019). In sum, Study 1 provided evidence supporting

the content validity of the English version of FFNI.

Chinese Version FFNI

We also established the content validity of the Chinese version of FFNI. First, we translated the original English FFNI into the Chinese version, following the best-practice recommendations for scale translation in organizational research (Brislin, 1970; Heggstad et al., 2019; Klotz et al., 2023).⁴ Second, we collected a separate sample to test the content validity of the Chinese FFNI. We recruited 120 Chinese employees through *Credamo*, an online platform to collect Chinese samples. The *Credamo* studies have been published in management and psychology journals, such as *Journal of Applied Psychology* and *Journal of Management* (Hu et al., 2025; Ren et al., 2023). We paid the participants ¥2 (around \$0.3) for participating in this 8-minute survey. To ensure data quality, participants were excluded from the analysis if they failed to pass the practice trial. The final sample of 100 participants consisted of 67% females and ranged in age from 21 to 60 years old ($M = 32.61$, $SD = 7.75$). Their average job tenure was 8.87 years ($SD = 7.60$).

We utilized Anderson and Gerbing (1991)'s classic sorting approach to establish content validity, following Colquitt et al. (2019)'s detailed and replicable instructions. Specifically, we asked the participants to read the items (the presented order was random) and sort them into the appropriate fundamental follower need.

We evaluated the content validity by calculating two indices. First, to evaluate the

⁴ The scale translation involved a 4-stage translation process and a 5-member review team. First, we let *ChatGPT* work as a translation expert to do the initial translation; we gained Chinese FFNI v1. Second, Yiming ZHU, an expert in cross-cultural and evolutionary psychology, reviewed the v1. The first author revised the Chinese items according to Yiming's feedback and suggestions, gaining v2. Third, Wen ZHANG (an expert in English, psychology, and leadership), Yanying ZHANG (a master student majoring in public management and psychology), and Yu HAN (a layperson majoring in philosophy) reviewed v2 and gave feedback. Last, Rui DUAN (a layperson working in a construction and real estate company as a company executive) and the first author discussed and revised the items together to get the final Chinese FFNI, based on the review team's feedback and the back-translation items generated by *DeepSeek*.

definitional correspondence, we generated the index of the proportion of substantive agreement for each item ($p_{sa} = n_c/N$; n_c is the number who sorted the item correctly). Second, to evaluate the definitional distinctiveness, we generated the index of the substantive validity coefficient for each item ($c_{sv} = n_c - n_0 / N$; n_0 is the maximum number of times an item was sorted into any other follower need). p_{sa} or c_{sv} achieves the value 1 when all participants sort an item correctly. For the 22 items, the results showed that the average p_{sa} was .93 (ranging from .80 to 1.00), and the average c_{sv} was .88 (ranging from .61 to 1.00; see Table S3.1 in the SM for the full results).⁵ These two content validation indices confirmed that the Chinese FFNI items were equipped with *very strong* correspondence and distinctiveness, according to the evaluation criteria recommended by Colquitt et al. (2019).

Study 2: Psychometric Properties

Study 2 had two goals. Most importantly, we examined the basic measurement properties and the multidimensional structure of FFNI across U.S., U.K., and Chinese samples, providing participants with either generic instructions (“*the leaders in your life*”; Samples A and B) or domain-specific instructions pertaining to the politics domain (“*the leaders in your country*”; Sample C) or work domain (“*the leaders in your workplace*”; Samples C, D, and E in two time points; quantitative data).⁶ Second, we investigated the specific types of leaders that individuals took into consideration when contemplating their needs as followers (qualitative data).

We generally expected that the 22-item FFNI to demonstrate a robust multidimensional structure across domains, countries, and times, such that a five-factor model representing the fundamental follower need for *protection*, *affiliation*, *status*, *guidance*, and *fairness* provided a

⁵ The results remained substantially unchanged when all available participants were included. The average p_{sa} was .91, and the average c_{sv} was .87.

⁶ We propose that Fundamental Follower Need (FFN) is a formative construct. That is, the observed indicators (*protection*, *affiliation*, *status*, *guidance*, *fairness*) define the latent construct FFN.

good fit to the data across samples.

Participants and Procedure

We recruited U.S. and U.K. samples (Samples A, B, C, and D) via *Prolific* based on the same prescreen requirements as in Study 1. Samples C and D included an extra requirement—full-time employment. We recruited the Chinese sample (Sample E) through *Credamo*. To ensure data quality, we excluded participants who failed to pass any of the careless response questions (Huang et al., 2012; Meade & Craig, 2012). We also excluded Sample E's participants with inconsistent demographic data between the *Credamo* platform, Time 1, and Time 2 responses.

Samples A and B. The final Samples A and B consisted of 203 and 1035 U.S. participants, respectively (2 and 19 participants were excluded from the initial Samples A and B, respectively). For Sample A, the participants were 48.3% female and ranged in age from 18 to 77 years old ($M = 35.71$, $SD = 13.19$). For Sample B, the participants were 48.0% female, and ranged in age from 18 to 85 years old ($M = 41.51$, $SD = 13.76$). The generic version of FFNI was administered to Samples A and B to test the measurement model using both relatively small and large sample sizes. For Sample A, we also asked participants what specific kinds of leaders they took into consideration immediately after they finished the generic FFNI.

Sample C. The final Sample C consisted of 328 U.S. participants after 12 participants were excluded based on the criteria. The participants were 49.4% female and ranged in age from 19 to 78 years old ($M = 40.61$, $SD = 14.27$). Both work and politics versions of FFNI were administered to Sample C. The presenting order of these two versions was random. We asked participants what specific kinds of leaders they took into consideration immediately after they finished the work- or politics-version FFNI.

Sample D. The final Sample D consisted of 295 U.K. participants after 11 participants were excluded based on the criteria. In this final sample, the gender balance was achieved (148 females, 50.17%; 145 males, 49.15%; and two others, 0.67%), while age spanned 18 to 74 ($M = 40.46$, $SD = 10.32$). The work version of FFNI was administered to Sample D.

Sample E. We recruited 400 participants via *Credamo* to get at least 300 participants after the two-wave data collection. At Time 1, the sample size was 396 after 4 participants were excluded based on the criteria. After one week, we collected the second wave data. 369 participants responded to this survey, and 2 participants were excluded based on the criteria. The final Sample E consisted of 367 Chinese participants. In this final sample, the gender balance was achieved (183 females, 49.86%; 184 males, 50.14%), while age spanned 20 to 65 ($M = 32.85$, $SD = 8.50$). The work version of FFNI was administered to Sample E at both time points.

Measures

Fundamental Follower Needs. We used the FFNI to measure followers' needs from their leaders. The participants responded to the 22 randomly presented items by indicating agreement on a 7-point scale with anchors of 1 = *strongly disagree* to 7 = *strongly agree*. Table 4 shows the content of the items. Crucially, we developed three versions of instructions for participants: generic, politics, and work. As mentioned, the generic version was administered to Samples A and B, the politics version was administered to Sample C, and the work version was administered to Samples C, D, and E.

Generic Version (US; Samples A and B). In the generic version, we asked participants to think about "*the people who you consider to be the leaders in your life – those individuals who influence you and other members of groups you belong to because they help you satisfy important needs and achieve important goals. They could be influential people in your family,*

community, society, team, or work organization.”

Politics Version (US; Sample C). In the politics version, we asked participants to think about “*the people who you consider to be the leaders in your COUNTRY (e.g., national politicians, local representatives) – those individuals who influence you and your fellow citizens because they help you satisfy important needs and achieve important goals.*”

Work Version (US, UK, CN; Sample C, D, and E). In the work version, we asked participants to think about “*the people who you consider to be the leaders in your WORK ORGANIZATION (e.g., senior managers, supervisors) – those individuals who influence you and your co-workers because they help you satisfy important needs and achieve important goals.*”

Leader Type. We asked participants to specify the role(s) of the leader(s) by text (e.g., work supervisor; the governor).

Results

CFA. Based on the recommendations for construct validity (Lambert & Newman, 2023; pp. 590-591), we conducted confirmatory factor analyses (CFA) to test the oblique five-factor measurement model following the conceptualization of fundamental follower needs. Tables 4 and 5 show the loadings and fit statistics of CFAs for the generic, work, and politics versions of FFNI across U.S., U.K., and Chinese samples. The CFA results indicated that the five-factor model—*protection, affiliation, status, guidance, and fairness*—had an acceptable fit for the generic and work versions but a poor fit for the politics version, whereas the six-factor model—separating *guidance* into *vision* and *expertise*—had a better level of fit overall and an acceptable fit across different domains, countries, and times (see Table 5).⁷

⁷ To explore whether alternative modeling approaches could provide additional insights into the factor structure, we also conducted exploratory factor analysis (EFA), exploratory structural equation modeling (ESEM), bifactor CFA, and bifactor ESEM using the largest available sample (i.e., Sample B), given the complexity of the models. Bifactor models yielded poorer fit, indicating that a general factor was not appropriate. Although ESEM, which integrates

Reliability. Tables 6 and 7 show the descriptive statistics, reliabilities, and correlation matrices of the generic, work, and politics versions of FFNI across U.S., U.K., and Chinese samples. Following the recommendations for reliability (Cortina et al., 2020; Revelle & Condon, 2019), we reported both Cronbach's α and model-based estimates ω , which ranged from .60 to .90.⁸ Overall, the results suggested the good reliability of the six-factor FFNI.

Leader Type. Figure 2 depicts the word clouds of frequently occurring leaders (see Table S4.4 in the SM for the detailed frequencies). When participants received the generic version of instruction (i.e., “*the leaders in your life*”), they considered various leaders when contemplating their needs as followers, such as leaders in family (e.g., father, mother, husband), community (e.g., coach), education (e.g., professor), religion (e.g., pastor), country (e.g., president, governor), and workplace (e.g., manager, supervisor). Specifically, leaders in politics and work were most frequently mentioned (see Figure 2a). When participants received the politics version of instructions (i.e., “*the leaders in your country*”), they considered different leaders, such as the president and mayor. When participants received the work version of instructions (i.e., “*the leaders in your workplace*”), they also considered leaders at different levels, such as the CEO and direct manager (see Figure 2b and 2c).

Conclusion

The results of Study 2 indicate that FFNI can provide a psychometrically sound measure of six fundamental follower needs—*protection*, *affiliation*, *status*, *vision*, *expertise*, and *fairness*—across domains, countries, and times. Specifically, according to the results, we split the broader

features of both EFA and CFA, can improve model fit, we prioritized the more restrictive CFA models for clarity and parsimony. Across all analyses, the six-factor model structure remained substantially unchanged (See S4 in the SM).

⁸ One exception was the reliability of FFNI for *vision* in the Chinese sample ($\alpha = .43$ and $.44$ at the two time points), which showed that although *vision* has relatively low alpha reliabilities, it does not seem to have major validity repercussions (see similar results in De Vries, 2013).

dimension of *guidance* into two factors: *vision*-based guidance and *expertise*-based guidance.

Study 3: Convergent and Discriminant Validity

The preliminary studies, Studies 1 and 2, developed a structural model of fundamental follower needs, as well as a needs inventory—the FFNI—to assess these needs. Building on this work, Study 3 pursued two primary goals. Specifically, we investigated both the convergent and discriminant validity of fundamental follower needs, following the approach used by George et al. (2023, pp.1961-1962). From a theoretical standpoint, FFNs are related to, but conceptually distinct from, several existing constructs (see the section on *Differentiating FFNs From Related Constructs*). Accordingly, we expected FFNs to show meaningful correlations with the need for leadership, fundamental social motives, and basic personal values, while remaining empirically separable from them. In contrast, FFNs were not expected to correlate with basic psychological needs satisfaction and frustration.

Participants and Procedure

Sample B. This sample encompassed the same cohort of U.S. adults utilized in Study 2, Sample B ($N = 1035$, 48.0% female, $M_{\text{age}} = 41.51$, $SD_{\text{age}} = 13.76$). To alleviate the participant burden, we randomly assigned them into four groups for distinct convergent, discriminant, and subsequent nomological network measures. Each group's sample size surpassed the recommended minimum of 250 observations for obtaining stable correlation estimates (Schonbrodt & Perugini, 2013). After providing demographic details (e.g., gender and age), participants completed the generic version FFNI, followed by group-specific measures. The first group ($N = 257$, 50.2% female, $M_{\text{age}} = 40.79$, $SD_{\text{age}} = 14.08$) responded to 53 items encompassing the need for leadership and fundamental social motives. The second group ($N = 260$, 47.7% female, $M_{\text{age}} = 42.45$, $SD_{\text{age}} = 13.38$) addressed 54 items related to basic values and

personality. The third group ($N = 257$, 46.3% female, $M_{\text{age}} = 40.28$, $SD_{\text{age}} = 13.55$) completed 56 items, including basic psychological needs satisfaction and frustration, affect, and intolerance of uncertainty. The fourth group ($N = 261$, 47.9% female, $M_{\text{age}} = 42.52$, $SD_{\text{age}} = 13.97$) engaged with a 51-item measure of implicit leadership theories (this measure was used for the subsequent nomological network validation). Participants were paid £0.70 (around \$0.86) for their participation in this 6-minute survey.

Sample D. This sample encompassed the same cohort of U.K. employees utilized in Study 2, Sample D ($N = 295$, 50.2% female, $M_{\text{age}} = 40.46$, $SD_{\text{age}} = 10.32$). Participants completed 73 items, including the work version FFNI, need for leadership, fundamental social motives, basic psychological needs satisfaction and frustration, and implicit leadership theories (this measure was used for the subsequent nomological network validation). Participants were paid £1.50 (around \$1.85) for their participation in this 14-minute survey.

Measures

Need for Leadership. We measured the need for leadership using De Vries et al. (2002)'s unidimensional scale ($\alpha = .92$). Participants rated each of the 17 items on a 5-point scale (1 = *not at all*, 5 = *a lot*). Example items included "I need my leader to decide what work should be done" and "I need my leader to provide me with information."

Fundamental Social Motives. The fundamental social motives inventory (Neel et al., 2016) was composed of six motives with six items each: self-protection, disease avoidance, affiliation (group), affiliation (exclusion concern), affiliation (independence), and status.⁹ Participants rated 36 items on a 7-point scale (1 = *strongly disagree*, 7 = *strongly agree*).

⁹ The fundamental social motives inventory in its original form incorporates motives pertinent to mate and kin relationships (namely, mate seeking, breakup concern, mate retention, kin care-family, kin care-children). We omitted these particular motives from our measurements because they do not align with the focus of the present study.

Example items included “I think a lot about how to stay safe from dangerous people” (self-protection) and “I do not like being at the bottom of a hierarchy” (status). The Cronbach’s α for the six scales ranged from .80 to .90.

Basic Values. We used the Revised Portrait Value Questionnaire (PVQ-RR; Schwartz & Cieciuch, 2022) to measure ten basic values with three items each: security-personal, security-social, benevolence-care, benevolence-dependability, achievement, power-resources, self-direction-thought, self-direction-action, universalism-concern, universalism-tolerance.

Participants rated 30 items on a 6-point scale (1 = *not like me at all*, 7 = *very much like me*).

Example items included “It is important to her/him that her/his country protect itself against all threats” (security-social) and “It is important to her/him to have the power that money can bring” (power-resources). The Cronbach’s α for the ten scales ranged from .59 to .90.

Basic Psychological Needs Satisfaction and Frustration. The Basic Psychological Need Satisfaction and Frustration Scale-General (BPNSNF; Chen et al., 2015; Van der Kaap-Deeder et al., 2020) was composed of six subscales with four items each: autonomy frustration, autonomy satisfaction, competence frustration, competence satisfaction, relatedness frustration, relatedness satisfaction. Participants rated 24 items on a 5-point scale (1 = *not true at all*, 5 = *completely true*). Example items included “I feel the relationships I have are just superficial” (relatedness frustration) and “I experience a warm feeling with the people I spend time with” (relatedness satisfaction). The Cronbach’s α for the six scales ranged from .61 to .94.

Results and Discussion

Convergent Validity. Convergent validity refers to how closely linked the focal construct is to conceptually related constructs. To establish convergent validity, we examined the correlations between the six types of fundamental follower needs and the need for leadership,

fundamental social motives, and basic personal values. As shown in Table 8, the unidimensional need for leadership was positively related to all six follower needs. The bivariate correlations between follower needs and their corresponding related fundamental social motives and basic personal values were positive. For instance, the fundamental follower need for *protection* was positively related to fundamental social motives of self-protection and disease avoidance, as well as basic values of security. Similarly, the fundamental follower need for *status* was positively associated with the fundamental social motive of status and basic values of achievement and power. In sum, the correlations between fundamental follower needs and their theoretically related constructs were positive and significant, providing evidence for convergent validity.

Contrary to our theoretical expectation, we observed several correlations between FFNs and basic psychological needs satisfaction. One possible explanation is that followers may be reluctant to display frustration toward their leaders, as expressing such dissatisfaction could signal personal inadequacy or weakness. When followers feel frustrated with the basic psychological needs, they may therefore prefer to seek need fulfillment in other social relationships (e.g., close friends) rather than from their leaders, leading to dampened FFNs. By contrast, when followers experience satisfaction of these basic needs, they may feel more confident and comfortable expressing their follower needs toward leaders, turn to leaders for fulfillment of these needs, and experience stronger FFNs, leading to positive associations between FFNs and basic needs satisfaction. Another interpretation is reverse causality: individuals with stronger FFNs may be more likely to experience satisfaction of these basic needs, as they actively seek leader support to fulfill them. We also found that correlations were stronger in Sample D, which used the workplace version of the FFNI (“leaders in your workplace”), compared to Sample B, which used the generic version (“leaders in your life”).

Drawing on leader distance theory (Antonakis & Atwater, 2002), proximal workplace leaders exert more direct and frequent influence on followers, making the link more salient. By contrast, distal leaders (e.g., president) are physically and psychologically farther away and rarely interact directly with followers, leading to weaker associations. This domain difference in leader distance likely contributed to the discrepancies in correlation strength.

Discriminant Validity. Discriminant validity refers to whether the focal construct is empirically distinct from theoretically different constructs. We utilized the CI_{CFA} technique to examine the factor correlations between fundamental follower needs and need for leadership, fundamental social motives, basic personal values, and basic psychological needs satisfaction and frustration for the purpose of confirming discriminant validity (Rönkkö & Cho, 2022). We expected these correlations to be “low enough for the measures to be regarded as measuring distinct constructs.” Specifically, we estimated CFA models that included fundamental follower needs inventory and the discriminant measure (e.g., a seven-factor model with six follower needs and the need for leadership), scaling the latent variables by fixing their variances to 1. Next, we inspected the upper limits (UL) of the 95% CIs of the estimated factor correlations and compared their values against .90 (.9 < UL < 1 indicated moderate problem, 1 < UL indicated severe problem). The results showed that all the ULs were less than this cutoff, providing evidence of the generic and work versions of FFNI’s discriminant validity (see S5 in the SM for the full results).

Conclusion

The results of Study 3 support both convergent and discriminant validity of FFNI by testing their relationships with the need for leadership, fundamental social motives, basic personal values, and basic psychological needs satisfaction and frustration.

Study 4: Antecedents of Fundamental Follower Needs

In this study, we mapped the nomological network of follower needs by testing their associations with the antecedents about “*where*” (e.g., leadership domains and countries) and “*who*” (stable individual differences; e.g., age and gender). We generally expected that the fundamental follower needs would vary across leadership domains (work and politics), countries (U.S., U.K., China), age, and gender, with the exact patterns of these differences remaining an open question for empirical investigation.

Participants

We utilized Sample C to explore the effect of leadership domains (politics vs. work) on the six fundamental follower needs. Focusing on the work domain, we used Samples C, D, and E to investigate the effect of countries (U.S., U.K., and China) on the fundamental follower needs. Last, we analyzed the data from Samples A to E to examine the effects of age and gender.¹⁰

Results and Discussion

Domains. We compared followers’ needs from their workplace and country leaders by paired T-test (i.e., work and politics domains; see Table 9).¹¹ First, followers needed more *protection* from their country’s leaders than workplace leaders—a finding consistent with our anticipation, given that physical and psychological safety concerns are typically more pronounced at the societal level than in the workplace. Second, followers needed more *affiliation, status, expertise, and fairness* from their workplace leaders compared to national

¹⁰ In Sample B, we also preliminarily explored the associations between other potential antecedents (e.g., personality traits, affect, and intolerance of uncertainty) and the fundamental follower needs. See S9 in the SM for the full results.

¹¹ Before comparing the mean values, we have examined the measurement invariance for domains, countries, times, and gender (Cheung & Rensvold, 2002). See S6 in the SM for the full results. As a robustness check, we also conducted MAN(C)OVA analyses to examine the effects of domains, countries, age, and gender on FFNs, with age and gender entered simultaneously as predictors. MAN(C)OVA (multivariate analysis of covariance) is advantageous because it accounts for intercorrelations among dependent variables. The robustness checks yielded results consistent with those reported in the main text. Detailed results are available in S8 of the SM.

leaders, likely because such needs are more readily addressed in the proximate context of organizational leadership. Third, leaders' guidance based on *vision* did not differ significantly, indicating the equal importance in both domains.

Countries. Focusing on the work domain, we compared fundamental follower needs across different countries using one-way ANOVA (i.e., the U.S., U.K., and China; see Figure 3). The results showed significant variations in all fundamental follower needs, except for *expertise*, implying that the importance of leader competence remains universal. Two other findings emerged. First, though sharing Western cultural roots, U.K. followers reported generally lower needs from their workplace leaders than U.S. followers (*affiliation, status, vision, and fairness, p* < .05; *protection and expertise, n.s.*). Second, consistent with our anticipation and prior cross-cultural research (Pick et al., 2022), the FFN for *status* was significantly higher in the Chinese sample than in the U.S. and U.K. ($p < .001$), a pattern aligning with the intensity of resource competition in high-density populations (Sng et al., 2018).

Age. Table 10 shows the correlations between age and fundamental follower needs. Four key findings emerged. First, consistent with our anticipation, older followers generally expressed less need for *expertise* from leaders because they could rely on their own accumulated experience and skills, though the effects were tiny. Second, younger followers showed a generically higher FFN for *status* than older ones, but this effect diminished when narrowing down to the politics and work domains. Third, age was only positively associated with the FFN for *protection* in the politics domain but not in the work domain (interaction of age and domain: $F(1, 324) = 6.21, p = .013$, partial $\eta^2 = 0.02$; see Figure S7 in the SM). That is, older followers only needed more *protection* from their country's leaders, but not workplace leaders. Fourth, age was only negatively related to the FFN for *fairness* in the Chinese sample but not in other

samples. This result might be explained by two points. The Chinese sample was younger on average ($M = 32.85$, $SD = 8.5$, $Max = 65$) than the U.S. and U.K. samples ($M = 35.71$ - 41.51 , $SD = 10.32$ - 14.27 , $Max = 74$ - 85). Besides, a supplementary analysis revealed a U-shape curvilinear relationship in China ($b = 0.07$, $se = 0.02$, $p < .001$), with *fairness* FFN increasing after age 41.35.¹² This pattern by and large aligned with China's "curse of 35" phenomenon (Li, 2023), where age discrimination peaks around mid-career. The resurgence of *fairness* FFN among older Chinese employees may reflect their direct experiences with workplace ageism.

Gender. Figure 4 and Table 11 show the T-test and mini meta-analysis results examining gender differences in fundamental follower needs (Goh et al., 2016). Across domains and countries, men had a higher FFN for *status*, while women showed higher FFNs for all other dimensions (*protection*, *affiliation*, *vision*, *expertise*, and *fairness*). Notably, these effects varied depending on specific domains. For instance, similar to age-related patterns, the gender gap in generic FFN for *status* became non-significant when narrowing down to the politics and work domains. That is, both female and male followers equally valued leaders who could enhance their status and wealth in organizational and political contexts. Besides, the FFNs for *protection* and *vision* showed more pronounced gender differences in the politics domain than in the work domain. Meanwhile, the FFN for *expertise* demonstrated a significant gender difference in the work domain but not in the politics domain. *Fairness* and *affiliation* FFNs maintained consistent gender differences across domains, suggesting that women constantly needed more *fairness* and *affiliation* from their leaders.

Conclusion

The findings from Study 4 are generally aligned with our expectations, revealing variations

¹² This age was calculated from the equation: $fairness = 0.07 (Zscore_age)^2 - 0.14 (Zscore_age) + 6.17$.

in fundamental follower needs vary across leadership domains (work and politics), countries (U.S., U.K., and China), age, and gender.

Study 5: Outcomes of Fundamental Follower Needs

In this study, we mapped the nomological network of follower needs by testing their associations with the outcomes about “*cognition*” (e.g., implicit leadership theories) and “*perception*” (e.g., perceived leadership effectiveness). Study 5 had three main goals. First, we demonstrated the incremental validity of certain FFN dimensions over others and the concurrent criterion validity using the implicit leadership theories as criterion variables (US; Sample B). Second, we established the incremental validity relative to prior needs or social motives measures (UK; Sample D). Third, using a non-Western sample we provided evidence for the predictive criterion validity using perceived leadership effectiveness as the criterion variables (CN; Sample E). We generally expected that the fundamental follower needs would relate meaningfully with implicit leadership theories and perceived leadership effectiveness, and would also provide unique information beyond existing measures of needs.

Incremental Validity of Certain FFN Dimensions Over Others

Participants. We analyzed the data from Sample B to examine the incremental validity of certain FFN dimension over others and the concurrent criterion validity using the implicit leadership theories as criterion variables.

Implicit Leadership Theories. The Implicit Theories of Leadership (Offermann & Coats, 2018) comprised eleven dimensions: sensitivity, dedication, tyranny, charisma, strength, creativity, well-groomed, masculinity, intelligence, femininity, and ethics.¹³ Participants rated 51

¹³ As Offermann and Coats (2018, pp. 518-519) did in their study, besides the 46 items of the formal Implicit Leadership Theories scale, we added the Femininity dimension (*female, feminine*) to correspond to the Masculinity dimension, as well as three items related to the Ethics dimension (*fair, honest, ethical*).

items on a 10-point scale (1 = *not at all characteristic of a leader*, 10 = *extremely characteristic of a leader*). The Cronbach's α for the eleven scales ranged from .76 to .92.

Results and Discussion. Following a multidimensional approach (Lebreton et al., 2007), we conducted both hierarchical multiple regression and relative weights analysis (RWA; Tonidandel et al., 2009; Tonidandel & LeBreton, 2011, 2015) to complementally establish the incremental validity (see Table 12 and Figure 5). Regression analyses demonstrated each need added unique variance to the prediction of implicit leadership theories over and above the other FFNs. For instance, the FFN for *protection* explained significantly incremental variance for the Strength dimension of ILTs ($\Delta R^2 = .03, p = .004$). RWA further revealed *protection* had the largest weight (.06), accounting for 35.35% of the total R^2 . Similarly, the FFN for *status* explained incremental variance for Tyranny ($\Delta R^2 = .06, p < .001$), Masculinity ($\Delta R^2 = .02, p = .019$), and Well-Groomed ($\Delta R^2 = .05, p < .001$), with RWA confirming its dominant influence on these dimensions. Notably, the above ILT dimensions—Strength, Tyranny, Masculinity, Well-groomed—aligned with the dominance-based leadership features outlined by the dual model of leadership and hierarchy (Cheng et al., 2013; Van Vugt & Smith, 2019). In contrast, prestige-based dimensions—Sensitivity, Dedication, Charisma, Creativity, Intelligence, and Ethics—were mainly explained by the FFN for *affiliation*, *vision*, *expertise*, and *fairness*. Despite strong correlations among these needs (r ranged from .60 to .72; see Table 7), hierarchical regression confirmed their incremental contributions to those prestige-based dimensions. *Vision* explained incremental variance for Sensitivity ($\Delta R^2 = .01, p = .044$), Dedication ($\Delta R^2 = .03, p = .001$), Charisma ($\Delta R^2 = .01, p = .049$), and Creativity ($\Delta R^2 = .02, p = .018$). *Expertise* explained incremental variance for Sensitivity ($\Delta R^2 = .04, p < .001$), Charisma ($\Delta R^2 = .02, p = .008$), and Intelligence ($\Delta R^2 = .02, p = .018$). *Fairness* explained incremental

variance for Dedication ($\Delta R^2 = .03, p < .001$), Intelligence ($\Delta R^2 = .03, p = .003$), and Ethics ($\Delta R^2 = .04, p < .001$) over other five follower needs. These findings underscored the complementary roles of our two theoretical foundations—the updated Pyramid of Needs (*affiliation*; Kenrick et al., 2010) and the Triad Model of Follower Needs (*vision, expertise, fairness*; De Waal-Andrews & Van Vugt, 2020). By integrating these perspectives, our FFN framework provided a more nuanced understanding of how distinct follower needs shape leadership prototypes.

Incremental Validity Relative to Prior Needs Measures

Participants. We used Sample D to establish the incremental validity relative to prior needs or social motives measures.

Measures. We measured FFNI-work, Need for Leadership, Fundamental Social Motives, Basic Psychological Needs Satisfaction and Frustration, and Implicit Leadership Theories. The instruments had been introduced previously in Study 3.

Results and Discussion. Table 13 shows the results of hierarchical multiple regressions. As indicated by the change in R^2 values, FFNI added unique variance to the prediction of all dimensions of implicit leadership theories over and above the need for leadership, fundamental social motives, and basic psychological needs satisfaction and frustration, regardless of whether the three measures entered the regression separately (Model A, B, C) or together (Model D in Table 13). The consistent results across models suggested that the FFN framework provided distinct and meaningful insights into leadership prototypes beyond established needs and social motivational frameworks.

Predictive Criterion Validity

Participants. We analyzed the data from Sample E to examine the predictive validity of FFNI for perceived leadership effectiveness. At Time 1, the work version of FFNI was

administered. After one week at Time 2, the measures of perceived leadership effectiveness were administered.

Perceived Leadership Effectiveness. The measures were mainly adapted from transformational leadership, paternalistic leadership, and status attainment framework (see Table S10.1 in the SM for the detailed items). To conserve space, the sources of all the measures are presented in the Appendix; all items have the validated Chinese version. The instructions for participants were as follows: “*We will now present approximately 50 leadership descriptions. We are interested in how well these descriptions match your expectations of effective leadership at this moment in time. These descriptions are about subordinates’ expectations of effective leadership in a Work Organization (e.g., senior managers, direct supervisors). Please answer to what extent you perceive these leadership descriptions as effective if these individuals were your own leaders. For each description, indicate your perceptions of its effectiveness (1 = not at all effective as a leader, 7 = extremely effective as a leader).*”

Results and Discussion. As shown in Table 14, our pre-registered hypotheses received partial empirical support. Hierarchical linear regressions demonstrated robust associations between specific FFNs and corresponding leadership dimensions regarding their effectiveness.¹⁴ Specifically, the FFN for *affiliation* was positively associated with the perceived effectiveness of leaders’ benevolence (Cheng et al., 2014; Cheng et al., 2004) and team-building behaviors (Wang & Howell, 2010). The FFN for *vision* was positively associated with the perceived effectiveness of leaders’ vision communication behaviors (Wang & Howell, 2010). The FFN for *expertise* was positively associated with the perceived effectiveness of leaders’ intellectual

¹⁴ For correlations among dependent variables, see S11 in the SM. For transparency, we also reported the results of structural equation modeling in Table 14, as pre-registered. In addition, given the correlations among dependent variables, we conducted path analyses as robustness checks, simultaneously entering all FFNs and dependent variables into the models. The results remain substantially unchanged; see S12 in the SM for detailed results.

stimulation behaviors (Wang & Howell, 2010). The FFN for *fairness* was positively associated with the perceived effectiveness of leaders' moral character (Cheng et al., 2014; Cheng et al., 2004). These relationships remained significant even after controlling for all other FFNs. These results provided support for Hypotheses 2, 4, 5, and 6, suggesting that followers' needs shaped their evaluations of leader effectiveness in theoretically predictable ways.

Contrary to expectations, the FFNs for *protection* and *status* failed to predict perceived effectiveness of dominance-based leadership styles (e.g., Authoritarianism, Narcissism, Dominance), despite their established relationships with dominance-based implicit leadership theories (Strength, Tyranny, Masculinity, Well-groomed; as shown in Sample B). This intriguing discrepancy suggested that while FFNs for *protection* and *status* may influence leadership prototypes, they did not necessarily translate to effectiveness perceptions. Besides, this finding pointed to a promising avenue for future studies—exploring the role of situations and events (see subsequent Direction for Future Research section). For instance, would inter-group conflict situations (e.g., trade wars, organizational crises) strengthen the FFN *protection*-authoritarianism link?

Conclusion

The findings from Study 5 provide evidence for the incremental validity and predictive criterion validity of FFNI. The results demonstrate that certain FFN dimensions account for unique variance not only relative to other fundamental follower needs but also beyond established needs constructs in predicting implicit leadership theories. Besides, the FFNI shows relationships between follower needs and corresponding perceptions of leadership effectiveness.

General Discussion

Our research addressed four central questions concerning the development,

conceptualization, and assessment of fundamental follower needs (FFNs). First, we argued that a taxonomy of follower needs, grounded in evolutionary theory, is essential because existing followership models, while valuable, lack a coherent explanation for why followers differ in their expectations and evaluations of leaders. Traditional models often classify followers by types (e.g., Kelley, 1988) or roles (e.g., Kellerman, 2007), but focus on observable behaviors rather than the deeper psychological mechanisms that drive them. In contrast, an evolutionary perspective offers a unifying and functional logic: follower needs are domain-specific motivational systems shaped by recurring coordination problems in ancestral environments, such as threat defense, resource allocation, and conflict management. This approach identifies which needs are fundamental (e.g., protection, affiliation, status) and why they are likely to generalize across domains, cultures, histories, and individuals. Crucially, it also shifts the analytical lens from leader-centric theories to a follower-driven understanding of leadership, with practical implications for research and development programs.

Second, we established a comprehensive conceptual framework to define and organize FFNs. Drawing on the Triad Model of Follower Needs (De Waal-Andrews & Van Vugt, 2020) and the Revised Pyramid of Needs (Kenrick et al., 2010), and guided by our empirical findings, we identified six needs—*protection*, *affiliation*, *status*, *vision*, *expertise*, and *fairness*—that reflect core adaptive problems that leaders have historically helped to solve. Crucially, our framework advances the Triad Model by further differentiating guidance into two distinct forms: *vision*-based and *expertise*-based. This refinement allows for more precise analysis; for instance, our data revealed that gender was primarily associated with the FFN for *vision*, whereas domain differences were more pronounced for *expertise*. Combining these dimensions into a single “guidance” construct would have masked such important nuances (see Figure S7.2 in the SM).

More broadly, these six needs are firmly rooted in the evolutionary behavioral sciences—spanning psychology, anthropology, and behavioral biology—and offer a unifying explanatory framework for understanding leadership endorsement across diverse contexts (Von Rueden & Van Vugt, 2015). Without this differentiation, we risk missing important nuances in how follower needs and, relatedly, leadership ideals vary across domains, national cultures, and demographics—differences well documented in studies like the GLOBE project (Aktas et al., 2016; House et al., 2004; Van de Vliert, 2006). Our taxonomy advances beyond earlier follower typologies (Kellerman, 2007; Kelley, 1988) by distinguishing followers not by types or roles, but by the specific problems they seek leaders to resolve (Bastardo & Adriaensen, 2023; Lonati & Van Vugt, 2024). This functionally and evolutionarily grounded taxonomy provides a robust framework for understanding the diversity of followership.

Third, we developed and validated a measurement tool, the 22-item Fundamental Follower Needs Inventory (FFNI). The FFNI demonstrates strong content validity (Study 1), has a robust multidimensional factor structure (Study 2), and sound convergent and discriminant validity (Study 3). It also differs with contextual and individual variation (Study 4) and shows incremental predictive validity for leadership ideals and perceived leader effectiveness beyond other need dimensions (Study 5). Although scholars frequently reference follower needs to explain leader preferences (Sergent & Stajkovic, 2020; Wang et al., 2023), they rarely assess these needs directly. In this regard, our contribution extends beyond the Triad Model of Follower Needs (De Waal-Andrews & Van Vugt, 2020), which articulated a compelling yet preliminary theoretical framework but explicitly called for the development of empirical tools to assess these needs. The FFNI answers this call by offering a practical tool for systematically exploring how stable traits and situational cues relate to leadership preferences and evaluations, thereby

enabling more rigorous and cumulative empirical research on followership.

Finally, we examined the antecedents and outcomes of FFNs through a preliminary mapping of their nomological network, using an input-process-output framework (see Figure 1). We found that FFNs vary with individual characteristics (e.g., age, gender) and contextual factors (e.g., domains, cultures). For example, women reported higher FFNs for *affiliation* than men across generic, workplace, and political domains—consistent with women’s generally stronger preferences for emotional and social support (Hill, 1987). In terms of outcomes, different FFNs predicted different leadership ideals. Individuals high in *protection* and *status* needs were more likely to favor strong, dominant leaders, while those high in *vision*, *expertise*, and *fairness* needs endorsed leaders who were competent, principled, and prestige-based. Contextual variation was also evident: the *protection* need was more salient in politics than in the workplace, likely due to the greater stakes involved in national security compared to organizational life. These findings suggest that FFNs are important psychological drivers of followers’ implicit leadership theories, shaping both their ideals and their evaluations of leader effectiveness.

Directions for Future Research

Testing the Mediating Role of Follower Needs

Fundamental follower needs are central to an input-process-output framework (Lewis et al., 2017), where contextual factors (e.g., culture, economic uncertainty) and individual differences (e.g., personality traits) influence specific follower needs, which in turn shape leadership-related outcomes (e.g., leadership ideals, perceived effectiveness). While prior research has shown, for example, that intergroup conflict increases preference for dominant leaders (Laustsen & Petersen, 2017; Laustsen et al., 2025; Spisak et al., 2012), and that individuals high in honesty-

humility are less favorable toward autocratic leaders (Dinic et al., 2023), these studies often assume yet do not empirically test the mediating role of follower needs—likely due to a lack of validated measures. The FFNI fills this gap, enabling researchers to identify which needs are activated in a given context and how they mediate the effects of personal or situational factors on leadership evaluations. Future research should also go beyond cognitive and perceptual outcomes to examine behavioral consequences of follower needs, such as voting in elections, leader support, resistance, or insubordination in organizational settings.

Cultural Generalizability Versus Specificity of Follower Needs

Another key direction is to examine how FFNs vary across cultures. Our nomological network suggests that national culture can influence both the activation and salience of specific follower needs (Lonati & Van Vugt, 2024). Just as researchers have documented cross-cultural patterns in fundamental social motives (Pick et al., 2022), similar efforts could investigate how ecological and cultural variables shape follower needs. For instance, does high population density increase the salience of the status follower need? Do aggressive wars elevate status needs, while defensive wars activate protection needs? Could the prominence of paternalistic leadership in East Asia (Cheng et al., 2014; Pellegrini & Scandura, 2008; Wang et al., 2018; Wang et al., 2023) be partly explained by a culturally elevated FFN for status and protection? Beyond national cultures, organizational ecologies may also matter. In hospitals, for instance, threats from infectious diseases may heighten the need for protection, while in universities, environments prioritizing inquiry and innovation may elevate needs for expertise and vision—shaping the kinds of leadership styles that emerge and succeed in each setting.

Exploring Causal Effects of Contextual Cues

While our studies focused on relatively stable antecedents of fundamental follower needs—

the “who” and “where” of them—future research should examine temporally dynamic antecedents—the “when” question. Experimental vignette studies offer a useful method for manipulating contextual cues that reflect evolutionarily recurrent challenges and thereby activate specific follower needs. For example, intergroup conflict may heighten the FFN for protection (Laustsen & Petersen, 2017), economic inequality may increase the FFNs for fairness and status (Long et al., 2024; Sprong et al., 2019), and male-biased sex ratios may intensify the FFN for protection and status (Sheng et al., 2024). These manipulations can test whether and how follower needs are activated in response to time-sensitive or event-driven social and ecological changes.

Fluctuations in Fundamental Follower Needs

Future research should examine how fundamental follower needs fluctuate over time. Consistent with our conceptualization, FFNs are expected to display a degree of trait-like stability while remaining sensitive to situational cues and changing environmental demands. Methods such as experience sampling or daily diary studies could effectively capture these short-term variations in follower needs (Baer et al., 2022) and identify their proximal triggers, such as daily experiences of uncertainty (Tepper et al., 2018). Investigating these temporal dynamics would offer valuable insight into the state-like flexibility of FFNs and how they adapt to shifting social and contextual conditions.

Interactions Between Situations and Individual Differences

Future studies should also explore person $\times$ situation interactions in the activation of FFNs. For example, individuals with a prevention-focused coping style (Lockwood et al., 2002) may exhibit a stronger need for protection in response to threats like pandemics, while promotion-focused individuals may focus more on acquiring status through their leader. Similarly, people

high in openness to experience may be more likely to activate needs for vision and expertise—especially when exposed to innovation-driven contexts such as the rise of artificial intelligence (von Stumm, 2018). Understanding how traits and environments jointly shape follower needs will help refine predictions about leader evaluations across varying contexts.

Limitations

While our research offers a theoretically grounded and empirically validated framework for understanding fundamental follower needs, several limitations should be acknowledged.

First, our cross-cultural scope, although spanning North America (United States), Europe (United Kingdom), and East Asia (China), remains incomplete. We did not include participants from other major cultural regions, such as South Asia, Latin America, Africa, or the Middle East. Given the possibility that ecological and cultural factors shape the salience of specific follower needs, future studies should replicate and extend our findings in underrepresented regions to test the broader cross-cultural generalizability of the FFN framework.

Second, all data were collected through online platforms (*Prolific* and *Credamo*). While these are reputable and increasingly accepted alternatives to traditional online panels (Douglas et al., 2023; Palan & Schitter, 2018; Peer et al., 2017), they still rely on self-selected samples and may underrepresent certain populations, such as older adults, non-internet users, or individuals in lower socioeconomic groups. To enhance ecological validity, future research should complement online sampling with field studies in organizational, political, or educational contexts, where follower-leader dynamics unfold in real-time.

Third, our research relied entirely on self-report measures and correlational designs, limiting causal inference. Although we established associations between FFNs and various antecedents and outcomes, we cannot determine the directionality or underlying mechanisms.

For instance, while followers with high protection needs endorse strong leaders, it remains unclear whether perceived threats increase protection needs, or whether certain individuals are chronically inclined toward this need. Future studies should use experimental manipulations (e.g., vignettes or simulations of threat, inequality, or uncertainty) to test how specific environmental cues activate FFNs and influence leadership evaluations.

Fourth, we focused primarily on cognitive and perceptual outcomes (e.g., leadership ideals, perceived effectiveness), rather than behavioral consequences. Yet follower needs likely shape tangible behaviors such as support for political candidates, compliance or resistance in organizations, and even turnover intentions. Longitudinal or behavioral data—such as voting patterns, employee performance records, or real-world leader evaluations—would provide a more complete picture of how FFNs translate into action over time.

Practical Implications

Despite these limitations, our research offers several important practical implications for leaders, followers, and organizations. First, the FFNs help followers develop self-awareness about the types of leadership they are likely to seek, especially in response to personal or situational needs. For leaders, understanding FFNs can improve their ability to diagnose and respond to the diverse motivations of those they lead. One practical application is the use of the FFNI in structured performance reviews, where managers and employees discuss alignment between leadership style and individual needs.

Second, in leadership development programs, the FFNs framework provides a theory-driven tool for training managers to recognize and respond to variation in follower expectations. By helping leaders tailor their behaviors, such as offering guidance, enforcing fairness, or fostering affiliation, the FFNI can support more motivational and effective leadership practices.

Third, the framework also offers insights into why leaders may fail to attract or retain followers, whether in business or politics (Mackey et al., 2021; Van der Velde & Gerpott, 2023). In political contexts, FFNs can help explain voter preferences—such as support for dominant leaders during times of perceived threat or uncertainty—providing a more psychologically nuanced account of electoral behavior.

Finally, the FFNI's brevity and adaptability enhance its value as a practical assessment tool. Its efficient 22-item format can be easily tailored to various leadership domains (e.g., workplace, politics, and education) or leadership levels (e.g., direct supervisor, CEO) through simple modifications in item instructions. This makes the FFNI not only a valuable research instrument but also a scalable tool for leadership diagnostics and interventions in real-world settings.

Conclusion

This research presents and validates a comprehensive framework for understanding and measuring the fundamental follower needs. We identify six core needs—*protection*, *affiliation*, *status*, *vision*, *expertise*, and *fairness*—that capture the evolved psychological mechanisms explaining why people are drawn to certain types of leaders and leadership qualities. These needs are firmly grounded in evolutionary and psychological theory and are empirically linked to key contextual and individual factors, including leadership domain, national culture, age, gender, implicit leadership theories, and perceptions of leadership effectiveness. At the core of this framework is the Fundamental Follower Needs Inventory (FFNI)—a concise, 22-item tool providing a reliable and valid assessment of follower needs across contexts. By integrating theory and measurement, this work offers a unifying foundation for future leadership research and practice. Understanding what followers truly seek from leaders is vital for fostering more adaptive, effective, and humane leadership in today's complex world.

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Table 1*Proposed Fundamental Follower Needs Framework*

	Fundamental Follower Need	Adaptive coordination problem	Psychological adaptation	Leadership ideals	Famous example	Sample item from FFNI
	Protection	Physical and social threats to group	Averting threats	strong, tough	Winston Churchill	Having a leader who takes care of my security is valuable to me.
	Affiliation	Inclusion and social support	Building alliances	compassionate, friendly	Satya Nadella	I wish to have a leader who creates a good group atmosphere.
	Status	Power and social rank dynamics	Getting valuable resources	domineering, power-hungry	Jeff Bezos	I am motivated to follow a leader who helps me become wealthy.
	Guidance	Adaptation to change (e.g., food/water scarcity)				
	- Vision	Lack of understanding	Moving towards a desired goal	charismatic, courageous	Elon Musk	Having a leader who moves the group towards a shared direction is valuable to me.
	- Expertise	Lack of competencies	Learning valuable skills	educated, intelligent	Warren Buffett	I need a leader who enables me to learn from them.
	Fairness	Resource distribution	Resolving and managing intragroup conflicts	honest, ethical	Nelson Mandela	It's important for me to have a leader who manages within-group conflicts in a just way.

Note. The icons are designed by Freepik (<https://www.freepik.com/icons>).

Table 2*Overview of Study Phases and Samples*

Phases	Purposes	Methods Used	Domain	Sample	<i>N</i>
Preliminary	Item generation, selection, refinement		—	US, UK	952
Study 1	Content validation	Rating approach	—	US	205
		Sorting approach	—	CN	100
Study 2	Reliability & psychometric properties	Item reliability (α and ω), CFA	Generic	US (Sample A)	203
			Generic	US (Sample B)	1035
			Politics	US (Sample C)	328
			Work	US (Sample C)	328
			Work	UK (Sample D)	295
			Work	CN (Sample E; T1)	396
			Work	CN (Sample E; T2)	367
Study 3	Convergent and discriminant validity	Correlations, CI_{CFA}	Generic	US (Sample B)	1035
			Work	UK (Sample D)	295
Study 4	Antecedents	T-test, ANOVA, correlations	Generic	US (Sample A)	203
			Generic	US (Sample B)	1035
			Politics	US (Sample C)	328
			Work	US (Sample C)	328
			Work	UK (Sample D)	295
			Work	CN (Sample E; T1)	396
			Work	CN (Sample E; T2)	367
Study 5	Outcomes				
	- Incremental validity of certain FFN dimensions over others	Correlations, hierarchical multiple regression,	Generic	US (Sample B)	1035
	- Incremental validity relative to prior needs measures	relative weights analysis, structure equation modeling	Work	UK (Sample D)	295
	- Predictive criterion validity		Work	CN (Sample E; T1)	396
			Work	CN (Sample E; T2)	367

Note. FFN = Fundamental Follower Need. US = United States. UK = United Kingdom. CN = China. T1/T2 = Time 1/2. CFA = Confirmatory Factor Analysis. CI_{CFA} = 95% Confidential Intervals of the estimated CFA factor correlations.

Appendix*Overview of Measures Used in the Studies*

Category		Measure	N_{item}	Source	Studies and Phases							
					1	2	3		4	5		
					US/CN		US	UK		US	UK	CN
1.	Follower Needs	FFNI items	22	The present research	✓							
2.	Follower Needs	FFNI-Generic	22	The present research		✓	✓		✓	✓		
3.	Follower Needs	FFNI-Politics	22	The present research		✓			✓			
4.	Follower Needs	FFNI-Work	22	The present research		✓		✓	✓		✓	✓
5.	Related Constructs	Need for Leadership	17	De Vries et al. (2002)			✓	✓			✓	
6.	Related Constructs	Fundamental Social Motives	36	Neel et al. (2016)			✓	✓			✓	
7.	Related Constructs	Basic Personal Values	30	Schwartz and Cieciuch (2022)			✓					
8.	Related Constructs	Basic Psychological Needs	24	Chen et al. (2015)			✓	✓			✓	
9.	Antecedents	Age	1	—					✓			
10.	Antecedents	Gender	1	—					✓			
11.	Outcomes	Implicit Leadership Theories	51	Offermann and Coats (2018)						✓	✓	
12.	Outcomes	Leadership Effectiveness										✓
1)	Outcomes	Authoritarianism	5	Cheng et al. (2004; 2014)								✓
2)	Outcomes	Benevolence	5	Cheng et al. (2004; 2014)								✓
3)	Outcomes	Moral Character	6	Cheng et al. (2004; 2014)								✓
4)	Outcomes	Team-Building	5	Wang and Howell (2010)								✓
5)	Outcomes	Vision Communication	4	Wang and Howell (2010)								✓
6)	Outcomes	High-Expectation	5	Wang and Howell (2010)								✓
7)	Outcomes	Intellectual Stimulation	4	Wang and Howell (2010)								✓
8)	Outcomes	Dominance	4	Bai et al. (2020)								✓
9)	Outcomes	Competence	4	Bai et al. (2020)								✓
10)	Outcomes	Virtue	6	Bai et al. (2020)								✓
11)	Outcomes	Safety	3	X. W. Hu et al. (2025)								✓
12)	Outcomes	Narcissism	6	Back et al. (2013)								✓

Note. Study 1: Content validation. Study 2: Reliability & psychometric properties. Study 3: Convergent and discriminant validity. Study 4: Antecedents. Study 5: Criterion/predictive validity and incremental validity.

US = United States. UK = United Kingdom. CN = China. Studies 2 and 4 involve all samples from the U.S., U.K., and China.

Table 3*Content Validation Statistics for the Fundamental Follower Needs Inventory (Study 1)*

Focal Subscale				Orbiting Subscales								Validation	
Construct	M_1	M_2	M	Group 1				Group 2				htc^a	htd^b
				Orbit1	M	Orbit2	M	Orbit3	M	Orbit4	M		
Protection (P)	6.57	6.48	6.52	A	3.34	S	2.23	F	3.53	G	2.92	0.93	0.59
Affiliation (A)	6.43	6.15	6.29	S	2.74	G	3.99	P	3.98	F	4.58	0.90	0.41
Status (S)	6.45	6.20	6.32	G	3.90	F	2.60	A	3.18	P	3.01	0.90	0.53
Guidance (G)	5.87	5.68	5.78	F	3.71	P	2.81	S	3.34	A	3.97	0.83	0.39
Fairness (F)	6.29	6.27	6.28	P	3.23	A	5.02	G	3.75	S	2.73	0.90	0.43
Definition of the Focal Construct													
Protection	Need for leaders who ensure (groups of) followers remain safe, especially in the face of external threat.												
Affiliation	Need for leaders who ensure (groups of) followers feel strongly connected and supported.												
Status	Need for leaders who help (groups of) followers to acquire status and resources.												
Guidance	Need for leaders who provide guidance to (groups of) followers based on their vision and expertise.												
Fairness	Need for leaders who ensure (groups of) followers are treated fairly and who effectively manage conflicts of interest within the group.												

Note. $N = 205$ ($N_{\text{Group1}} = 103$, $N_{\text{Group2}} = 102$). Participants rated on a 7-point scale (1 = *Item does an EXTREMELY BAD job of measuring [the focal construct]*, 7 = *Item does an EXTREMELY GOOD job of measuring [the focal construct]*).

a. htc (Hinkin Tracey correspondence) = $M_{\text{focal}} / 7$, the maximum value is 1. $htc \geq .90$ could be interpreted as *Very Strong* definitional correspondence; $.82 \leq htc \leq .85$ could be interpreted as *Moderate* definitional correspondence.

b. htd (Hinkin Tracey distinctiveness) = Average of all $(M_{\text{focal}} - M_{\text{orbit}}) / (7 - 1)$, ranging from -1 to 1. $htd \geq .23$ could be interpreted as *Very Strong* definitional distinctiveness; based on Colquitt et al. (2019)'s correlation-based criteria when the average correlation between focal scale and orbiting scales is stronger (.51 or above).

Table 4*Final Items and Confirmatory Factor Analyses Loadings of the FFNI (Study 2)*

Follower Needs and Items		Generic	Politics	Work			
		US	US	US	UK	CN	
						T1	T2
Protection							
1	I wish to have a leader who positions themselves between my group and an outside threat.	.72/.77	.72	.79	.66	.52	.56
2	I am motivated to follow a leader who stands up for me against outsiders.	.76/.78	.75	.81	.72	.58	.61
3	Having a leader who takes care of my security is valuable to me.	.73/.74	.81	.75	.70	.64	.67
4	It's important for me to have a leader who protects the group from any harm.	.83/.82	.85	.78	.78	.51	.66
Affiliation							
5	I wish to have a leader who creates a good group atmosphere.	.82/.81	.83	.77	.69	.52	.45
6	I am motivated to follow a leader who creates a sense of camaraderie.	.64/.78	.81	.79	.73	.50	.46
7	Having a leader who can bring the group together is valuable to me.	.77/.82	.76	.79	.82	.57	.59
8	It's important for me to have a leader who makes me feel part of the group.	.73/.76	.74	.80	.73	.44	.52
Status							
9	I need a leader who improves my standing.	.84/.86	.80	.85	.83	.72	.74
10	I wish to have a leader who helps our group achieve status.	.83/.80	.79	.75	.73	.44	.47
11	I am motivated to follow a leader who helps me become wealthy.	.76/.73	.77	.73	.66	.42	.46
12	Having a leader who helps me achieve prominence is valuable to me.	.87/.86	.87	.88	.82	.59	.64
Guidance-Vision							
13	I wish to have a leader who sets clear goals.	.81/.79	.80	.76	.63	.38	.38
14	I am motivated to follow a leader who has a clear vision.	.72/.79	.81	.69	.69	.49	.48
15	Having a leader who moves the group towards a shared direction is valuable to me.	.79/.78	.72	.78	.78	.47	.47
Guidance-Expertise							
16	I need a leader who enables me to learn from them.	.77/.77	.87	.82	.73	.50	.64
17	Having a leader who is a good teacher is valuable to me.	.81/.82	.88	.82	.82	.61	.73
18	It's important for me to have a leader who offers advice.	.77/.74	.86	.73	.70	.57	.63
Fairness							
19	I need a leader who ensures that procedures are free of bias.	.63/.69	.69	.68	.65	.53	.43
20	I wish to have a leader who carefully listens to both parties when mediating a conflict.	.78/.80	.80	.77	.79	.49	.47
21	Having a leader who makes fair and balanced decisions is valuable to me.	.77/.83	.77	.68	.85	.49	.53
22	It's important for me to have a leader who manages within-group conflicts in a just way.	.66/.78	.78	.65	.72	.65	.54

Note. US = United States. UK = United Kingdom. CN = China. T1/T2 = Time 1/2. Loadings for the generic version are derived from Sample A ($N = 203$; the values left of each forward slash) and Sample B ($N = 1035$; the values right of each forward slash). Loadings for the politics version are derived from Sample C ($N = 328$). Loadings for the work version are derived from Sample C ($N = 328$; US), Sample D ($N = 295$; UK), and Sample E ($N_{T1} = 396$, $N_{T2} = 367$; CN).

Table 5*Fit Statistics for Confirmatory Factor Analyses of the FFNI (Study 2)*

Domains	Samples	Model	χ^2	df	CFI	TLI	RMSEA	SRMR	$\Delta\chi^2 (\Delta df)$	p
Generic	US	Sample A (N = 203)								
		Five-factor model	353.22	199	.94	.93	.04	.05	30.41 (5)	<.001
		Second-order factor model	331.25	197	.95	.94	.03	.05	8.44 (3)	.038
		Six-factor model	322.81	194	.95	.94	.03	.05		
Generic	US	Sample B (N = 1035)								
		Five-factor model	1034.52	199	.94	.93	.05	.04	259.89 (5)	<.001
		Second-order factor model	786.88	197	.96	.95	.04	.04	12.25 (3)	.007
		Six-factor model	774.63	194	.96	.95	.04	.04		
Politics	US	Sample C (N = 328)								
		Five-factor model	981.45	199	.83	.80	.08	.12	486.54 (5)	<.001
		Second-order factor model	596.89	197	.91	.90	.05	.08	101.98 (3)	<.001
		Six-factor model	494.91	194	.93	.92	.05	.06		
Work	US	Sample C (N = 328)								
		Five-factor model	536.45	199	.92	.91	.04	.05	102.88 (5)	<.001
		Second-order factor model	467.81	197	.94	.92	.03	.05	34.24 (3)	<.001
		Six-factor model	433.57	194	.94	.93	.03	.05		
Work	UK	Sample D (N = 295)								
		Five-factor model	500.01	199	.91	.89	.05	.06	81.38 (5)	<.001
		Second-order factor model	428.46	197	.93	.92	.04	.05	9.83 (3)	.020
		Six-factor model	418.63	194	.93	.92	.04	.05		
Work	CN	Sample E (N = 396; Time 1)								
		Five-factor model	324.51	199	.91	.90	.03	.05	18.13 (5)	.003
		Second-order factor model	307.75	197	.92	.91	.03	.05	1.37 (3)	.713
		Six-factor model	306.38	194	.92	.91	.03	.05		
Work	CN	Sample E (N = 367; Time 2)								
		Five-factor model	369.39	199	.87	.85	.04	.06	60.78 (5)	<.001
		Second-order factor model	308.61	197	.91	.90	.03	.05	3.07 (3)	.381
		Six-factor model	305.54	194	.91	.90	.03	.05		

Note. Five-factor model: *protection, affiliation, status, guidance, fairness*. Second-order factor model: Loading the first-order factors **guidance-vision** and **guidance-expertise** on the second-order factor **guidance**. Six-factor model: *protection, affiliation, status, guidance-vision, guidance-expertise, fairness*. CFI = Comparative Fit Index. TLI = Tucker-Lewis Index. RMSEA = Root-Mean-Square Error of Approximation. SRMR = Standardized Root Mean Square Residual. The models that fit better are bolded.

Table 6*Descriptive Statistics for the FFNI (Study 2)*

Follower Need	Generic		Politics	Work			
	US	US	US	US	UK	CN	CN
	(Sample A)	(Sample B)	(Sample C)	(Sample C)	(Sample D)	(Sample E) T1	(Sample E) T2
Protection	5.80 (0.95)	5.63 (1.04)	5.92 (0.94)	5.78 (0.96)	5.69 (0.83)	5.91 (0.63)	5.91 (0.68)
Affiliation	6.11 (0.77)	5.98 (0.85)	5.77 (0.98)	6.08 (0.85)	5.92 (0.79)	6.13 (0.53)	6.09 (0.53)
Status	4.99 (1.31)	4.68 (1.37)	4.87 (1.36)	5.19 (1.25)	4.93 (1.16)	5.79 (0.67)	5.78 (0.71)
Vision	6.07 (0.83)	6.02 (0.81)	6.10 (0.80)	6.17 (0.74)	6.00 (0.71)	6.19 (0.51)	6.18 (0.51)
Expertise	5.98 (0.89)	5.82 (0.92)	5.24 (1.33)	5.98 (0.90)	5.86 (0.86)	5.92 (0.71)	5.91 (0.81)
Fairness	6.25 (0.68)	6.26 (0.74)	6.17 (0.80)	6.28 (0.68)	6.14 (0.67)	6.25 (0.52)	6.23 (0.48)
Sample Size	203	1035	328	328	295	396	367

Note. US = United States. UK = United Kingdom. CN = China. T1/T2 = Time 1/2.

Table 7*Reliability Coefficients and Intercorrelations of the FFNI (Study 2)*

Domain	Follower Need	Reliability		Intercorrelations					
		α	ω	Protection	Affiliation	Status	Vision	Expertise	Fairness
Generic	<i>Sample A (US, N = 203; the values left of each forward slash)</i>								
	<i>Sample B (US, N = 1035; the values right of each forward slash)</i>								
	Protection	.84/.86	.85/.86	—	.71/.70	.55/.59	.64/.67	.70/.62	.67/.57
	Affiliation	.82/.87	.82/.87	.60/.60	—	.41/.42	.87/.85	.82/.78	.79/.81
	Status	.89/.88	.89/.88	.51/.52	.35/.38	—	.62/.49	.55/.53	.27/.23
	Vision	.82/.83	.82/.83	.53/.57	.71/.72	.52/.42	—	.86/.77	.67/.77
	Expertise	.82/.81	.82/.81	.59/.52	.67/.67	.49/.46	.71/.64	—	.74/.66
	Fairness	.80/.85	.79/.85	.54/.49	.60/.69	.24/.21	.52/.65	.58/.55	—
Politics	<i>Sample C (US, N = 328)</i>								
	Protection	.86	.86	—	.65	.52	.77	.43	.61
	Affiliation	.86	.86	.56	—	.52	.67	.71	.71
	Status	.88	.88	.48	.46	—	.39	.58	.24
	Vision	.81	.82	.65	.58	.35	—	.42	.82
	Expertise	.90	.90	.40	.65	.51	.38	—	.37
	Fairness	.84	.85	.50	.59	.21	.69	.32	—
Work	<i>Sample C (US, N = 328)</i>								
	Protection	.86	.86	—	.62	.58	.71	.61	.63
	Affiliation	.87	.87	.53	—	.49	.81	.83	.83
	Status	.88	.88	.53	.53	—	.51	.54	.35
	Vision	.79	.79	.58	.53	.45	—	.79	.85
	Expertise	.82	.83	.52	.58	.48	.64	—	.68
	Fairness	.79	.79	.49	.52	.30	.67	.55	—
	<i>Sample D (UK, N = 295)</i>								
	Protection	.80	.81	—	.45	.50	.34	.43	.54
	Affiliation	.83	.83	.68	—	.34	.27	.38	.40
	Status	.84	.84	.42	.43	—	.33	.53	.53
	Vision	.75	.75	.57	.65	.49	—	.34	.24
	Expertise	.80	.80	.52	.66	.38	.57	—	.44
	Fairness	.83	.84	.60	.62	.29	.68	.53	—

<i>Sample E (CN, T1, N = 396; the values left of each forward slash)</i>								
<i>Sample E (CN, T2, N = 367; the values right of each forward slash)</i>								
Protection	.65/.72	.66/.72	—	.37/.32	.47/.35	.38/.38	.29/.20	.61/.42
Affiliation	.61/.58	.60/.57	.24/.20	—	.25/.11	.80/.89	.64/.56	.69/.65
Status	.60/.66	.63/.68	.33/.28	.20/.09	—	.30/.30	.31/.32	.13/.11
Vision	.43/.41	.44/.42	.21/.21	.37/.43	.19/.18	—	.72/.46	.59/.51
Expertise	.61/.70	.62/.71	.20/.14	.35/.36	.22/.21	.34/.28	—	.51/.34
Fairness	.63/.56	.63/.56	.39/.26	.38/.36	.10/.08	.31/.27	.29/.22	—

Note. US = United States. UK = United Kingdom. CN = China. T1/T2 = Time 1/2.

The values in the lower triangle indicate Pearson's correlations, and those in the upper triangle indicate the CFA-based latent factor correlations. All correlations are significant at $p < .001$; two exceptions: (1) In Sample E (T1), p (*status* and *fairness*) $< .05$. (2) In Sample E (T2), p (*status* and *affiliation*) = .102, p (*status* and *fairness*) = .128.

Table 8*Correlations of the FFNI-Generic/Work with Convergent and Discriminant Measures (Study 3)*

Variables	Protection	Affiliation	Status	Vision	Expertise	Fairness
Need for Leadership	.43***/.31**	.52***/.39**	.44***/.44***	.47***/.40***	.51***/.42***	.44***/.27***
Fundamental Social Motives						
Self-Protection	.33***/.42***	.14*/.28***	.23***/.33***	.10/.24***	.14*/.21***	.13*/.22***
Disease Avoidance	.20**/.26***	.08/.05	.14*/.25***	.14*/.10	.17**/.12*	.12*/.11
Affiliation (Group)	.32***/.40***	.49***/.58***	.29***/.34***	.35***/.47***	.43***/.45***	.28***/.34***
Affiliation (Exclusion Concern)	.19**/.16**	.18**/.24***	.17**/.04	.03/.04	.14*/.13*	.04/.08
Affiliation (Independence)	-.06/-.05	-.09/-.19**	-.12/-.11	.12*/-.11	-.12/-.12*	.04/-.00
Status	.26***/.32***	.23***/.37***	.55***/.60***	.18**/.31***	.24***/.21***	.02/.17**
Basic Values						
Security-personal	.52***	.44***	.24***	.45***	.43***	.41***
Security-social	.34***	.28***	.18**	.33***	.26***	.25***
Benevolence-care	.27***	.34***	.14*	.40***	.28***	.37***
Benevolence-dependability	.36***	.47***	.19**	.48***	.44***	.39***
Achievement	.41***	.36***	.52***	.39***	.43***	.14*
Power-resources	.21***	.04	.52***	.05	.13*	-.10
Self-Direction-thought	.11	.20***	.03	.24***	.23***	.24***
Self-Direction-action	.21***	.27***	.05	.26***	.26***	.36***
Universalism-concern	.32***	.44***	.08	.42***	.39***	.57***
Universalism-tolerance	.33***	.47***	.18**	.45***	.45***	.51***
Basic Psychological Need Satisfaction and Frustration						
Autonomy Frustration	-.03/.03	-.00/-.08	.15*/.08	-.08/-.06	.02/-.04	-.08/-.07
Autonomy Satisfaction	.10/.25***	.06/.36***	.08/.32***	.20**/.32***	.08/.30***	.10/.28***
Competence Frustration	-.03/-.15*	.02/-.23***	.06/-.19**	-.11/-.15*	.09/-.12*	-.03/-.14*
Competence Satisfaction	.19**/.32***	.13*/.39***	.15*/.28***	.24***/.38***	.07/.22***	.20**/.40***
Relatedness Frustration	-.03/-.09	-.08/-.18**	.09/.02	-.14*/-.17**	-.00/-.11	-.16*/-.12*
Relatedness Satisfaction	.13*/.33***	.18**/.40***	.06/.22***	.22***/.35***	.08/.31***	.20**/.30***

Note. The values left of each forward slash are for Sample B (Generic FFNI, US; $N = 1035$); the values right of each forward slash are for Sample D (Work FFNI; UK; $N = 295$). Correlations between conceptually related constructs are bolded.

*. $p < .05$, **. $p < .01$, *** $p < .001$

Table 9*The Effect of Domains on Fundamental Follower Needs (Study 4)*

Follower Need	Politics Domain					Work Domain					Domain <i>d</i>	Domain <i>p</i>
	Women <i>M (SD)</i>	Men <i>M (SD)</i>	Combined <i>M (SD)</i>	Gender <i>d</i>	Gender <i>p</i>	Women <i>M (SD)</i>	Men <i>M (SD)</i>	Combined <i>M (SD)</i>	Gender <i>d</i>	Gender <i>p</i>		
Protection	6.05 (0.81)	5.82 (0.96)	5.94 (0.90)	.26	.022	5.91 (0.89)	5.66 (1.00)	5.79 (0.95)	.26	.021	-.20	<.001
Affiliation	5.90 (0.94)	5.66 (0.98)	5.78 (0.97)	.26	.022	6.19 (0.83)	5.97 (0.86)	6.08 (0.85)	.26	.019	.41	<.001
Status	4.79 (1.50)	4.98 (1.17)	4.88 (1.35)	-.14	.195	5.17 (1.31)	5.24 (1.17)	5.21 (1.24)	-.06	.605	.38	<.001
Vision	6.22 (0.75)	6.00 (0.84)	6.11 (0.80)	.27	.016	6.29 (0.67)	6.05 (0.79)	6.17 (0.74)	.33	.003	.10	.078
Expertise	5.32 (1.36)	5.21 (1.24)	5.26 (1.30)	.08	.448	6.08 (0.87)	5.90 (0.90)	5.99 (0.89)	.20	.075	.64	<.001
Fairness	6.31 (0.74)	6.02 (0.84)	6.17 (0.80)	.37	<.001	6.42 (0.60)	6.14 (0.72)	6.28 (0.68)	.42	<.001	.19	<.001
Sample Size	162	164	326			162	164	326				

Note. Gender *d* = Cohen's *d* for the mean-level difference between women and men, with positive values indicating higher scores for women. Domain *d* = Cohen's *d* for the mean-level difference between the politics and work domains, with positive values indicating higher scores in the work domain.

Table 10*The Correlations of Age and Fundamental Follower Needs (Study 4)*

Domain	Sample	<i>N</i>	<i>M</i> _{age} (<i>SD</i>)	Range _{age}	Protection	Affiliation	Status	Vision	Expertise	Fairness
Generic	US (Sample A)	203	35.71 (13.19)	18-77	.01	-.05	-.14*	-.04	-.10	-.09
Generic	US (Sample B)	1035	41.51 (13.76)	18-85	-.01	-.05	-.19***	-.00	-.11***	.04
Politics	US (Sample C)	328	40.61 (14.27)	19-78	.14**	.02	-.08	.07	-.01	.06
Work	US (Sample C)	328	40.61 (14.27)	19-78	.02	.06	-.09	.08	-.02	.00
Work	UK (Sample D)	295	40.46 (10.32)	18-74	-.00	-.01	-.13*	-.05	-.09	.03
Work	CN (Sample E; T1)	396	32.75 (8.35)	19-65	-.03	.04	.00	.06	-.03	-.13*
Work	CN (Sample E; T2)	367	32.85 (8.50)	20-65	-.04	.02	.03	.08	-.04	-.11*

Note. *. $p < .05$, **. $p < .01$, *** $p < .001$.

Table 11*The Summary of Effect Sizes of Gender on Fundamental Follower Needs (Study 4)*

Need	Sample	Domain		
		Generic	Politics	Work
Protection	Sample A (US; 98/102)	.07		
	Sample B (US; 497/527)	.19		
	Sample C (US; 162/164)		.26	.26
	Sample D (UK; 148/145)			.30
	Sample E (CN; 199/197) – Time 1			-.01
	Sample E (CN; 183/184) – Time 2			-.08
	Meta-analysis	.17	.26	.10
	95% CI	 [.06, .28]	 [.04, .48]	 [-.01, .20]
Affiliation	Sample A (US; 98/102)	.33		
	Sample B (US; 497/527)	.22		
	Sample C (US; 162/164)		.26	.26
	Sample D (UK; 148/145)			.36
	Sample E (CN; 199/197) – Time 1			.13
	Sample E (CN; 183/184) – Time 2			-.07
	Meta-analysis	.24	.26	.16
	95% CI	 [.13, .35]	 [.04, .48]	 [.05, .26]
Status	Sample A (US; 98/102)	-.50		
	Sample B (US; 497/527)	-.22		
	Sample C (US; 162/164)		-.14	-.06
	Sample D (UK; 148/145)			-.12
	Sample E (CN; 199/197) – Time 1			.09
	Sample E (CN; 183/184) – Time 2			-.00
	Meta-analysis	-.27	-.14	-.01
	95% CI	 [-.38, -.15]	 [-.36, .08]	 [-.12, .09]
Vision	Sample A (US; 98/102)	.02		
	Sample B (US; 497/527)	.08		
	Sample C (US; 162/164)		.27	.33

Expertise	Sample D (UK; 148/145)			.21
	Sample E (CN; 199/197) – Time 1			-.01
	Sample E (CN; 183/184) – Time 2			-.14
	Meta-analysis	.07	.27	.08
	95% CI	[-.04, .18]	 [.05, .49]	 [-.02, .19]
	Sample A (US; 98/102)	.07		
	Sample B (US; 497/527)	.06		
	Sample C (US; 162/164)		.08	.20
	Sample D (UK; 148/145)			.43
	Sample E (CN; 199/197) – Time 1			.13
Fairness	Sample E (CN; 183/184) – Time 2			.20
	Meta-analysis	.06	.08	.23
	95% CI	[-.05, .17]	 [-.14, .30]	 [.12, .33]
	Sample A (US; 98/102)	.26		
	Sample B (US; 497/527)	.22		
	Sample C (US; 162/164)		.37	.42
	Sample D (UK; 148/145)			.39
	Sample E (CN; 199/197) – Time 1			.14
	Sample E (CN; 183/184) – Time 2			.03
	Meta-analysis	.23	.37	.23
	95% CI	 [.11, .34]	 [.15, .59]	 [.12, .33]

Note. The numbers in the parentheses are the sample sizes of women/men, respectively. The effect size is Cohen's *d* for the mean-level difference between women and men, with positive values indicating higher scores for women.

Table 12*Incremental Validity and Relative Importance of Fundamental Follower Needs for Implicit Leadership Theories (Study 5; Sample B)*

Predictor	Correlation	Hierarchical Multiple Regression				Relative Weights Analysis		
	<i>r</i>	<i>b</i>	<i>SE</i>	β	ΔR^2	RW	[RW CI]	RW%
Criterion: Sensitivity ($R^2 = 0.26$)								
Intercept		2.58***	0.70					
Protection	.29***	-0.03	0.11	-0.02	0.00	0.02	[0.01, 0.05]	7.07
Affiliation	.42***	0.16	0.18	0.09	0.00	0.05	[0.02, 0.09]	19.73
Status	.18**	-0.11	0.07	-0.10	0.01	0.01	[0.00, 0.01]	2.80
Vision	.43***	0.33*	0.16	0.20	0.01*	0.06	[0.02, 0.11]	23.81
Expertise	.46***	0.49***	0.13	0.32	0.04***	0.09	[0.04, 0.15]	34.74
Fairness	.35***	0.00	0.16	0.00	0.00	0.03	[0.01, 0.07]	11.86
Criterion: Dedication ($R^2 = 0.39$)								
Intercept		3.14***	0.49					
Protection	.39***	0.04	0.08	0.04	0.00	0.04	[0.01, 0.08]	9.50
Affiliation	.54***	0.18	0.13	0.13	0.00	0.08	[0.03, 0.13]	20.40
Status	.19**	-0.08	0.05	-0.10	0.01	0.01	[0.00, 0.01]	1.84
Vision	.57***	0.38***	0.11	0.30	0.03***	0.11	[0.04, 0.18]	28.19
Expertise	.43***	0.02	0.09	0.02	0.00	0.04	[0.02, 0.08]	11.24
Fairness	.56***	0.39***	0.12	0.25	0.03***	0.11	[0.03, 0.20]	29.84
Criterion: Tyranny ($R^2 = 0.12$)								
Intercept		3.83***	1.02					
Protection	.20**	0.25	0.16	0.14	0.01	0.03	[0.01, 0.06]	21.16
Affiliation	.06	-0.11	0.27	-0.05	0.00	0.00	[0.00, 0.01]	2.90
Status	.31***	0.44***	0.11	0.32	0.06***	0.08	[0.03, 0.14]	63.99
Vision	.07	-0.10	0.24	-0.04	0.00	0.00	[0.00, 0.01]	3.15
Expertise	.08	-0.09	0.19	-0.05	0.00	0.00	[0.00, 0.01]	3.61
Fairness	-.03	-0.22	0.24	-0.08	0.00	0.01	[0.00, 0.03]	5.18
Criterion: Charisma ($R^2 = 0.29$)								
Intercept		2.72	0.64					
Protection	.37***	-0.01	0.10	-0.01	0.00	0.03	[0.01, 0.08]	10.11
Affiliation	.46***	0.19	0.17	0.12	0.00	0.06	[0.02, 0.11]	19.38
Status	.33***	0.08	0.07	0.09	0.00	0.03	[0.01, 0.08]	11.87

Vision	.47***	0.29*	0.15	0.19	0.01*	0.06	[0.02, 0.13]	22.10
Expertise	.48***	0.32**	0.12	0.22	0.02**	0.08	[0.04, 0.13]	26.46
Fairness	.36***	0.00	0.15	0.00	0.00	0.03	[0.01, 0.08]	10.09
Criterion: Strength ($R^2 = 0.17$)								
Intercept		3.05	0.77					
Protection	.38***	0.35**	0.12	0.26	0.03**	0.06	[0.02, 0.12]	35.35
Affiliation	.29***	-0.13	0.20	-0.07	0.00	0.02	[0.01, 0.04]	10.19
Status	.29***	0.11	0.08	0.10	0.01	0.03	[0.01, 0.08]	19.02
Vision	.32***	0.19	0.18	0.11	0.00	0.03	[0.01, 0.07]	17.8
Expertise	.26***	-0.02	0.14	-0.01	0.00	0.01	[0.00, 0.04]	7.46
Fairness	.26***	0.20	0.18	0.09	0.00	0.02	[0.01, 0.05]	10.18
Criterion: Creativity ($R^2 = 0.25$)								
Intercept		3.30***	0.62					
Protection	.35***	0.02	0.10	0.02	0.00	0.03	[0.01, 0.07]	11.60
Affiliation	.45***	0.24	0.17	0.16	0.01	0.06	[0.02, 0.11]	22.72
Status	.25***	-0.00	0.07	-0.00	0.00	0.01	[0.00, 0.05]	5.60
Vision	.46***	0.34*	0.14	0.24	0.02*	0.07	[0.02, 0.12]	27.47
Expertise	.41***	0.14	0.12	0.10	0.00	0.05	[0.02, 0.09]	18.64
Fairness	.37***	0.05	0.15	0.03	0.00	0.03	[0.01, 0.08]	13.97
Criterion: Well-Groomed ($R^2 = 0.15$)								
Intercept		2.58*	1.09					
Protection	.29***	0.19	0.17	0.10	0.00	0.03	[0.01, 0.08]	19.40
Affiliation	.21***	-0.14	0.29	-0.06	0.00	0.01	[0.00, 0.02]	5.54
Status	.37***	0.46***	0.12	0.30	0.05***	0.08	[0.03, 0.16]	55.1
Vision	.23***	0.11	0.25	0.05	0.00	0.01	[0.00, 0.04]	8.68
Expertise	.23***	0.06	0.21	0.03	0.00	0.01	[0.00, 0.04]	8.88
Fairness	.14*	0.04	0.26	0.01	0.00	0.00	[0.00, 0.01]	2.39
Criterion: Masculinity ($R^2 = 0.06$)								
Intercept		3.07**	0.98					
Protection	.15*	0.00	0.15	0.00	0.00	0.01	[0.00, 0.02]	11.24
Affiliation	.12	-0.12	0.26	-0.06	0.00	0.00	[0.00, 0.01]	4.93
Status	.22***	0.25*	0.10	0.19	0.02*	0.03	[0.00, 0.08]	55.08
Vision	.15*	0.18	0.23	0.09	0.00	0.01	[0.00, 0.03]	12.91

Expertise	.16*	0.10	0.18	0.05	0.00	0.01	[0.00, 0.03]	13.61
Fairness	.08	-0.04	0.23	-0.02	0.00	0.00	[0.00, 0.00]	2.23
Criterion: Intelligence ($R^2 = 0.18$)								
Intercept		4.20***	0.60					
Protection	.26***	0.08	0.09	0.08	0.00	0.02	[0.00, 0.05]	9.24
Affiliation	.30***	-0.27	0.16	-0.19	0.01	0.02	[0.01, 0.04]	10.27
Status	.17**	-0.03	0.06	-0.03	0.00	0.01	[0.00, 0.02]	3.33
Vision	.36***	0.18	0.14	0.14	0.01	0.04	[0.01, 0.08]	22.46
Expertise	.35***	0.27*	0.11	0.21	0.02*	0.04	[0.01, 0.08]	22.76
Fairness	.38***	0.44**	0.14	0.26	0.03**	0.06	[0.01, 0.13]	31.94
Criterion: Femininity ($R^2 = 0.02$)								
Intercept		4.85***	0.98					
Protection	-.06	-0.24	0.15	-0.15	0.01	0.01	[0.00, 0.03]	29.27
Affiliation	.04	0.35	0.26	0.16	0.01	0.00	[0.00, 0.02]	20.22
Status	-.03	-0.05	0.10	-0.04	0.00	0.00	[0.00, 0.01]	8.18
Vision	.00	0.01	0.23	0.00	0.00	0.00	[0.00, 0.00]	6.27
Expertise	.05	0.17	0.18	0.09	0.00	0.00	[0.00, 0.02]	23.49
Fairness	-.02	-0.28	0.23	-0.11	0.01	0.00	[0.00, 0.01]	12.57
Criterion: Ethics ($R^2 = 0.23$)								
Intercept		3.66***	0.65					
Protection	.18**	-0.12	0.10	-0.10	0.00	0.01	[0.00, 0.01]	3.91
Affiliation	.35***	0.04	0.17	0.03	0.00	0.04	[0.01, 0.08]	15.83
Status	.05	-0.11	0.07	-0.12	0.01	0.01	[0.00, 0.02]	2.40
Vision	.38***	0.29	0.15	0.19	0.01	0.05	[0.02, 0.11]	23.63
Expertise	.32***	0.17	0.12	0.12	0.01	0.03	[0.01, 0.07]	15.24
Fairness	.44***	0.53***	0.15	0.29	0.04***	0.09	[0.02, 0.17]	39.00

Note. $N = 261$. The incremental importance is indicated by the ΔR^2 (Hierarchical Multiple Regression). The bolded needs are the needs that explain a significant amount of variance in a certain dimension of implicit leadership theories (Relative Weights Analysis; bootstrapping = 10000, $\alpha = .05$).

RW = relative weight (within round error relative weights sum to R^2).

[RW CI] = 95% confidence interval of raw weight. Smaller CIs indicate greater precision.

RW% = relative weight rescaled as a percentage of predicted variance attributable to each predictor.

*, $p < .05$, **, $p < .01$, *** $p < .001$.

Table 13*Incremental Validity Relative to Prior Needs Measures for Implicit Leadership Theories (Study 5; Sample D)*

Criterion:	Model A		Model B		Model C		Model D	
<i>Sensitivity</i>	M1	M2	M1	M2	M1	M2	M1	M2
NFL	✓	✓					✓	✓
FSM			✓	✓			✓	✓
BPNSNF					✓	✓	✓	✓
FFNI		✓		✓		✓		✓
R^2	.12	.35	.21	.38	.22	.41	.30	.43
ΔR^2		.23		.17		.19		.13
p		<.001		<.001		<.001		<.001
Criterion:	Model A		Model B		Model C		Model D	
<i>Dedication</i>	M1	M2	M1	M2	M1	M2	M1	M2
NFL	✓	✓					✓	✓
FSM			✓	✓			✓	✓
BPNSNF					✓	✓	✓	✓
FFNI		✓		✓		✓		✓
R^2	.06	.36	.23	.38	.22	.40	.30	.41
ΔR^2		.30		.15		.18		.11
p		<.001		<.001		<.001		<.001
Criterion:	Model A		Model B		Model C		Model D	
<i>Tyranny</i>	M1	M2	M1	M2	M1	M2	M1	M2
NFL	✓	✓					✓	✓
FSM			✓	✓			✓	✓
BPNSNF					✓	✓	✓	✓
FFNI		✓		✓		✓		✓
R^2	.00	.09	.09	.14	.04	.13	.14	.19
ΔR^2		.09		.05		.09		.05
p		<.001		.015		<.001		.009

Criterion:	Model A		Model B		Model C		Model D	
<i>Charisma</i>	M1	M2	M1	M2	M1	M2	M1	M2
NFL	✓	✓					✓	✓
FSM			✓	✓			✓	✓
BPNSNF					✓	✓	✓	✓
FFNI		✓		✓		✓		✓
R^2	.14	.38	.25	.39	.22	.43	.34	.44
ΔR^2		.25		.14		.21		.10
p		<.001		<.001		<.001		<.001
Criterion:	Model A		Model B		Model C		Model D	
<i>Strength</i>	M1	M2	M1	M2	M1	M2	M1	M2
NFL	✓	✓					✓	✓
FSM			✓	✓			✓	✓
BPNSNF					✓	✓	✓	✓
FFNI		✓		✓		✓		✓
R^2	.01	.18	.13	.20	.07	.19	.15	.22
ΔR^2		.17		.07		.12		.07
p		<.001		<.001		<.001		<.001
Criterion:	Model A		Model B		Model C		Model D	
<i>Creativity</i>	M1	M2	M1	M2	M1	M2	M1	M2
NFL	✓	✓					✓	✓
FSM			✓	✓			✓	✓
BPNSNF					✓	✓	✓	✓
FFNI		✓		✓		✓		✓
R^2	.21	.45	.31	.45	.23	.46	.41	.49
ΔR^2		.23		.14		.23		.07
p		<.001		<.001		<.001		<.001
Criterion:	Model A		Model B		Model C		Model D	
<i>Well-Groomed</i>	M1	M2	M1	M2	M1	M2	M1	M2
NFL	✓	✓					✓	✓

FSM			✓	✓			✓	✓
BPNSNF					✓	✓	✓	✓
FFNI		✓		✓		✓		✓
R^2	.03	.15	.17	.21	.10	.20	.20	.25
ΔR^2		.13		.04		.10		.04
p		<.001		.021		<.001		.024
Criterion:	Model A		Model B		Model C		Model D	
<i>Masculinity</i>	M1	M2	M1	M2	M1	M2	M1	M2
NFL	✓	✓					✓	✓
FSM			✓	✓			✓	✓
BPNSNF					✓	✓	✓	✓
FFNI		✓		✓		✓		✓
R^2	.00	.11	.08	.16	.03	.14	.14	.21
ΔR^2		.10		.07		.11		.07
p		<.001		<.001		<.001		<.001
Criterion:	Model A		Model B		Model C		Model D	
<i>Intelligence</i>	M1	M2	M1	M2	M1	M2	M1	M2
NFL	✓	✓					✓	✓
FSM			✓	✓			✓	✓
BPNSNF					✓	✓	✓	✓
FFNI		✓		✓		✓		✓
R^2	.15	.37	.26	.38	.20	.39	.35	.42
ΔR^2		.22		.12		.19		.07
p		<.001		<.001		<.001		<.001
Criterion:	Model A		Model B		Model C		Model D	
<i>Feminity</i>	M1	M2	M1	M2	M1	M2	M1	M2
NFL	✓	✓					✓	✓
FSM			✓	✓			✓	✓
BPNSNF					✓	✓	✓	✓
FFNI		✓		✓		✓		✓

R^2	.01	.05	.03	.07	.02	.07	.05	.09
ΔR^2		.04		.04		.05		.04
p		.038		.073		.027		.049
Criterion:	Model A		Model B		Model C		Model D	
<i>Ethics</i>	M1	M2	M1	M2	M1	M2	M1	M2
NFL	✓	✓					✓	✓
FSM			✓	✓			✓	✓
BPNSNF					✓	✓	✓	✓
FFNI		✓		✓		✓		✓
R^2	.05	.25	.11	.28	.14	.29	.19	.32
ΔR^2		.21		.17		.15		.12
p		<.001		<.001		<.001		<.001

Note. $N = 295$. NFL = Need for Leadership. FSM = Fundamental Social Motives.

BPNSNF = Basic Psychological Needs Satisfaction and Frustration.

FFNI = Fundamental Follower Needs Inventory.

Table 14*Hypotheses and Linear Regression Results (Study 5; Sample E)*

No.	Follower Needs (IV; Time 1)	Leadership Effectiveness (DV; Time 2)	Rationale ^a	Framework Derived	Structural Equation Modeling		Linear Regression			
							Model 1		Model 2	
					β	p	β	p	β	p
1	Protection	a. Safety	Conceptual alignment	—	.43***	<.001	.20***	<.001	.05	.358
		b. Authoritarianism	Theory-based inference	PL	-.02	.803	-.01	.923	.01	.904
2	Affiliation	a. Benevolence	Conceptual alignment	PL	.68***	<.001	.31***	<.001	.10*	.048
		b. Team-building	Conceptual alignment	TL	.72***	<.001	.46***	<.001	.36***	<.001
3	Status	a. Narcissism	Theory-based inference	—	.04	.555	.03	.581	.04	.457
		b. Authoritarianism	Theory-based inference	PL	-.05	.471	-.02	.745	-.02	.672
		c. Dominance	Theory-based inference	SA	-.12	.089	-.05	.374	.02	.771
4	Vision	a. Vision communication	Conceptual alignment	TL	.58***	<.001	.32***	<.001	.18**	.001
		b. High-expectation	Conceptual alignment	TL	.59***	<.001	.28***	<.001	.10	.060
5	Expertise	a. Competence	Conceptual alignment	SA	.52***	<.001	.22***	<.001	.08	.161

		b. Intellectual stimulation	Conceptual alignment	TL	.68***	<.001	.31***	<.001	.13*	.015
6	Fairness	a. Moral character	Conceptual alignment	PL	.59***	<.001	.36***	<.001	.25***	<.001
		b. Virtue	Conceptual alignment	SA	.46***	<.001	.26***	<.001	.08	.136

Note. The pre-registered hypotheses are in the unified format (<https://aspredicted.org/vt3x-j9vb.pdf>): *The fundamental follower needs for [IV] will be positively associated with the perceived effectiveness of [DV].*

a. Two general types of rationale guide the hypothesis formulation: (1) Conceptual alignment: These hypotheses reflect a direct conceptual match between a specific follower need and the leadership styles that may fulfill it, which then could be perceived as more effective. For example, followers high in the FFN for *vision* are expected to perceive *vision communication* leadership as more effective. (2) Theory-based inference: These hypotheses are grounded in broader theoretical reasoning or prior empirical evidence, implying links between follower needs and leadership effectiveness. For example, followers with a strong FFN for *protection* may view authoritarian leaders as effective because such leaders convey aggression, decisiveness and control in managing outside threats. Similarly, followers high in the FFN for *status* may perceive narcissistic, dominant, or authoritarian leaders as more effective, as these leaders display confidence, assertiveness, and control that signal status and access to resources.

PL = Paternalistic Leadership Framework (Cheng et al., 2004; Cheng et al., 2014).

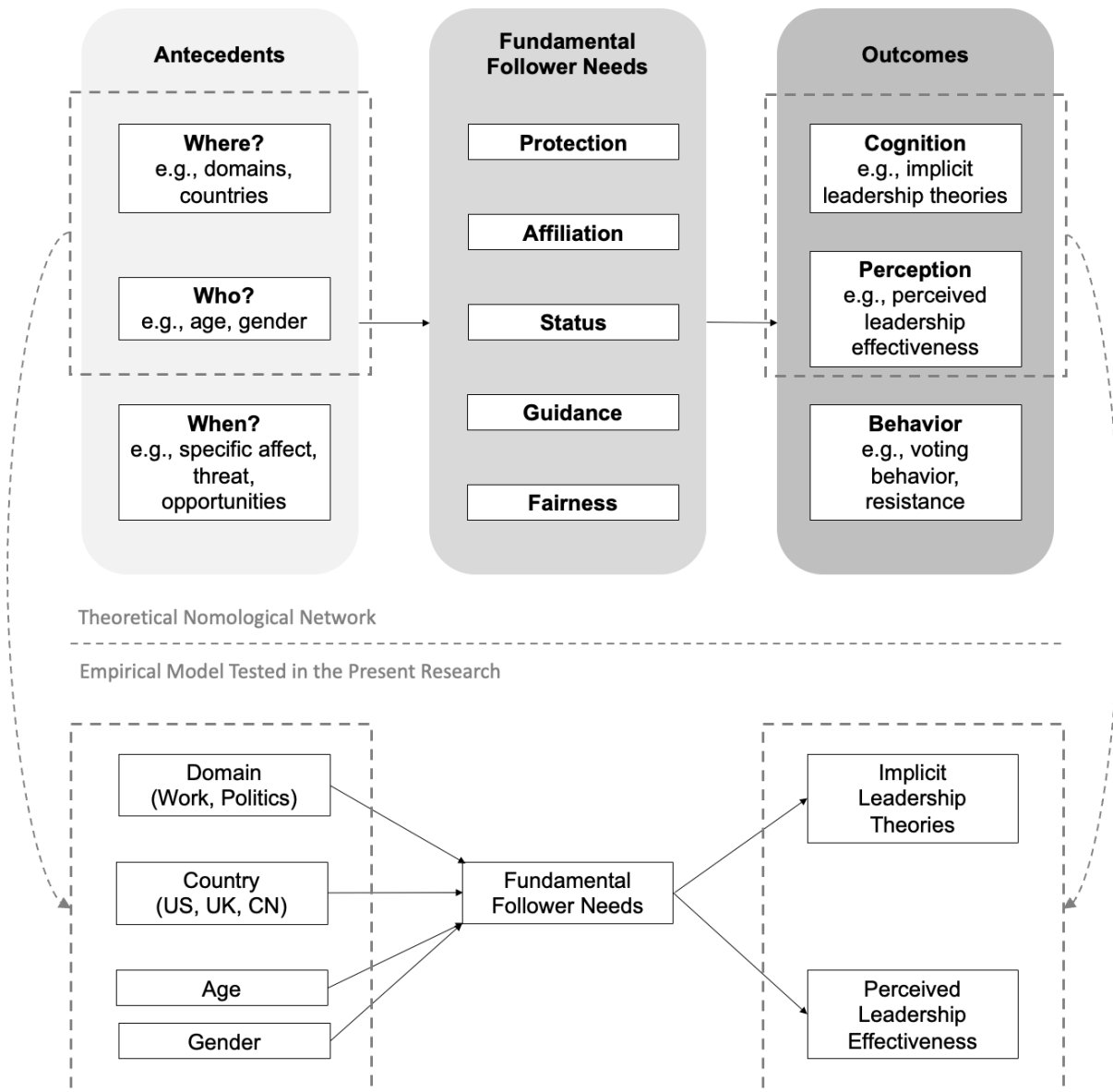
TL = Transformational Leadership Framework (Podsakoff et al., 1990; Wang & Howell, 2010).

SA = Status Attainment Framework (Bai et al., 2020).

Model 1: Linear regression models include the corresponding follower need.

Model 2: Linear regression models control for the other five follower needs.

*, $p < .05$, **, $p < .01$, ***, $p < .001$

Figure 1*Nomological Network of Fundamental Follower Needs*

Word Cloud of Frequently Occurring Leaders (Study 2)

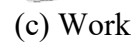
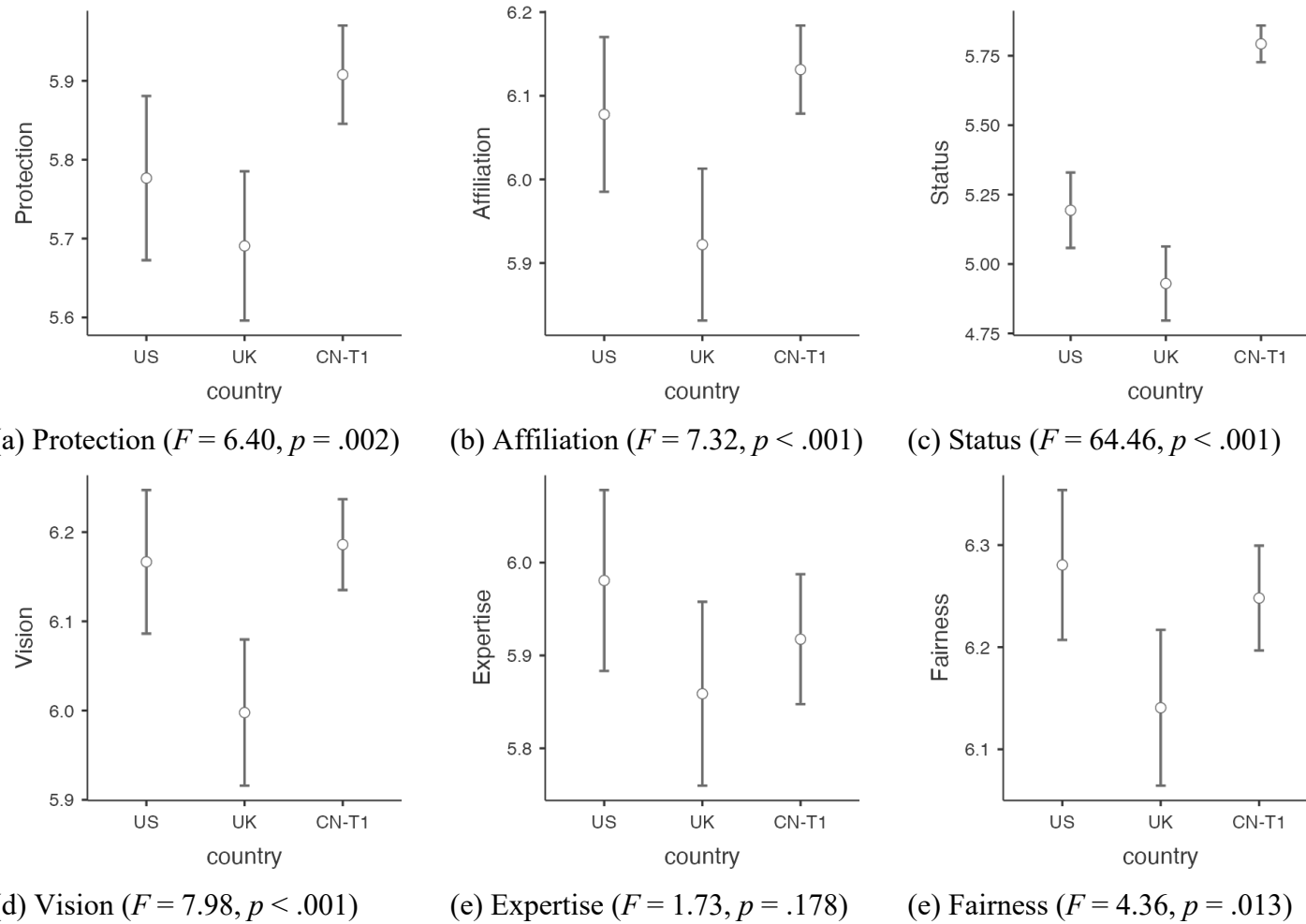
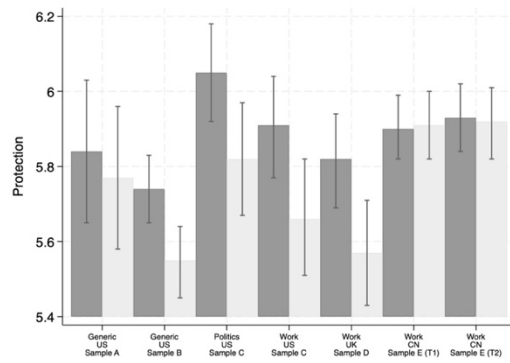
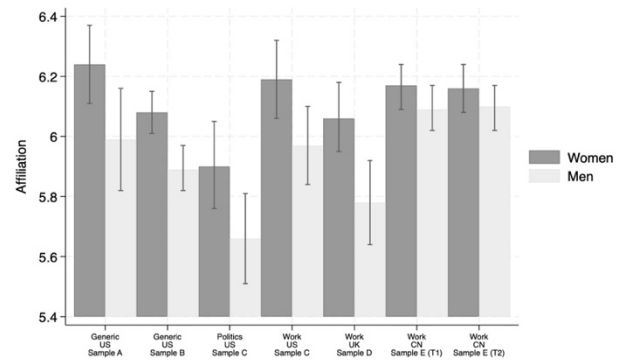
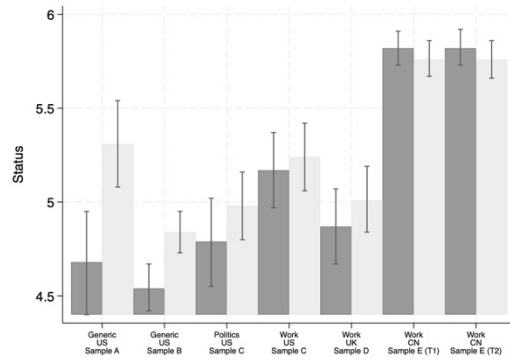
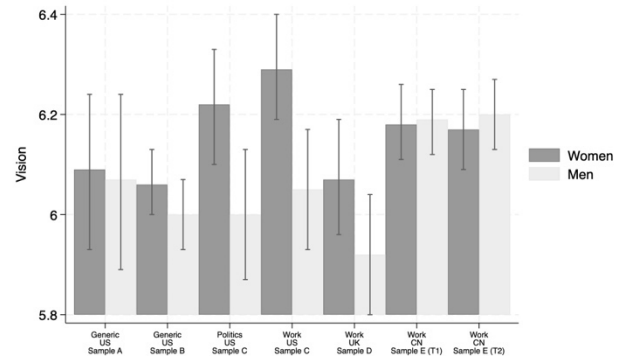
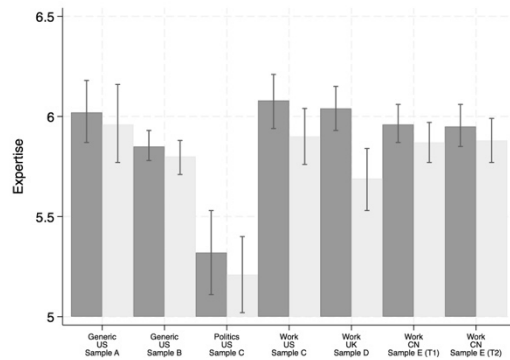
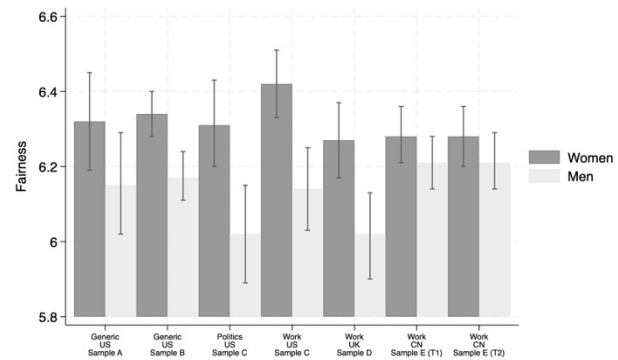


Figure 3*The Effect of Countries on the Fundamental Follower Needs (Study 4)*

Note. Error bars indicate 95% Confidence Intervals for the mean values. The values in the parentheses show the results of one-way ANOVA, $df_1 = 2, df_2 = 1016$.

Figure 4*The Effect of Gender on the Fundamental Follower Needs (Study 4)*(a) Protection ($d = .15$ [.08, .22])(b) Affiliation ($d = .20$ [.13, .27])(c) Status ($d = -.13$ [-.21, -.06])(d) Vision ($d = .10$ [.03, .17])(e) Expertise ($d = .14$ [.07, .21])(f) Fairness ($d = .24$ [.17, .32])

Note. The error bars indicate 95% Confidence Intervals (CIs) for the mean values. The values in the parentheses show Cohen's d and 95% CI from the mini meta-analysis across all samples (Goh et al., 2016), with positive values indicating higher scores for women.

Figure 5. Incremental Validity of Certain FFN dimensions over others (Study 5)